

●● OFFICIAL DOCUMENTATION

USER MANUAL

BIOMASS CALCULATOR

DETAILED GUIDE TO SOFTWARE OPERATION,
DATA INPUT METHODOLOGY & ANALYSIS

VERSION

1.0

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User Manual

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1) EULA

To use the program, you must agree to the End User License Agreement (EULA). Click **“Agree”** to continue to the **main menu**. Clicking **“Disagree”** will **close the Tool** immediately.

2) About

In the left-side column, click on the **purple-highlighted button [About]** to see the general information of the tool. Once the pop-up shows, click on the [X] button in the top left corner to go back to the Main Menu.

About the Tool

The tool calculates biomass for individual tree components—wood, bark, branches, and foliage—and ensures that the sum of these components equals the total aboveground biomass. It uses species-specific allometric models based on diameter at breast height (DBH) and, when available, tree height, offering two levels of precision:

DBH-based equations for situations where height data are unavailable.

DBH + height-based equations for improved accuracy when both measurements are provided.

Key features

- Covers 33 Canadian tree species, plus grouped equations for hardwoods, softwoods, and all species combined.
- Provides outputs suitable for forest carbon budget estimation, ecological modeling, and operational planning.

This tool is intended for researchers, forest managers, and policy analysts who require consistent and scientifically robust biomass estimates across Canada.

Development and contact

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3) Supported Species

The **green-highlighted button [Supported Species]** shows a look-up section where users can **find all tree species and associated species code numbers** that are currently **supported by this tool** for biomass calculations. Once the pop-up section shows up, **click on [Dismiss]** at the lower right corner of the section to **go back to the Main Menu**.

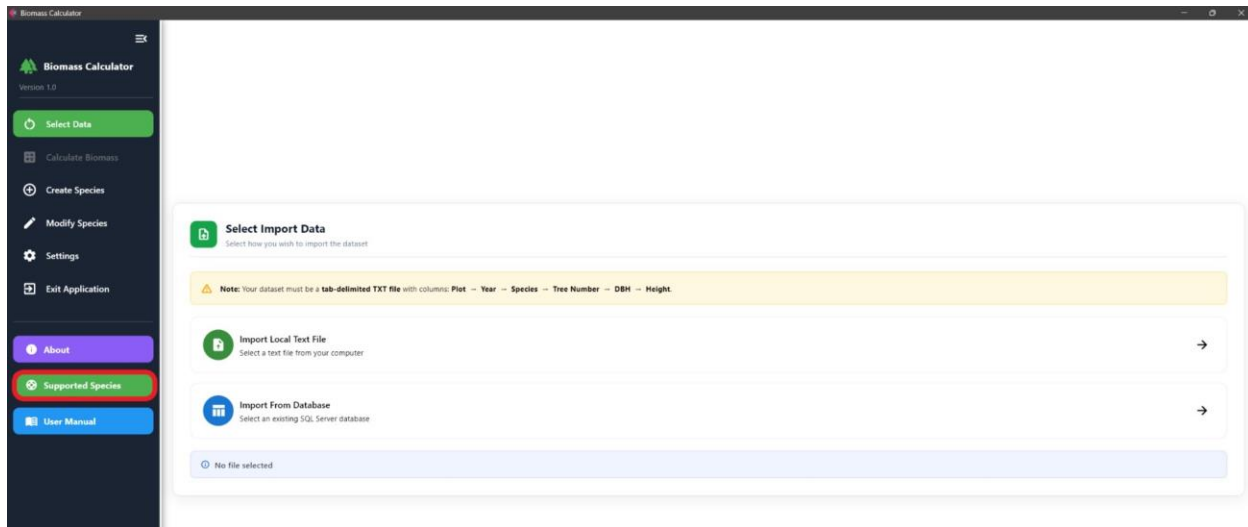


Figure 1. [Supported Species] button highlighted.

Within **[Supported Species]**, users can **search for species by name or code number**. If the user types of alphabet character(s), the function will find all matching species names that contain the string of characters (case insensitive). On the other hand, if the **user types a number**, the function will try to **find the species** that either has the **matching code number or contains that number in its code number**.

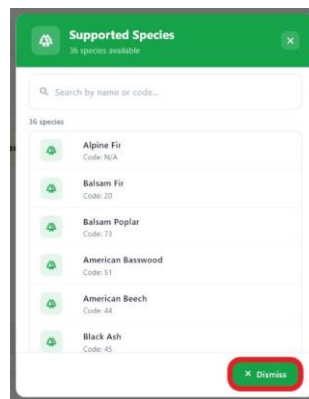


Figure 2. [Supported Species] section pop-up. [Dismiss] button highlighted

4) Select Data

To select the data to be used, open the “**Select Data**” menu by clicking on the Select Data button on the left sidebar.

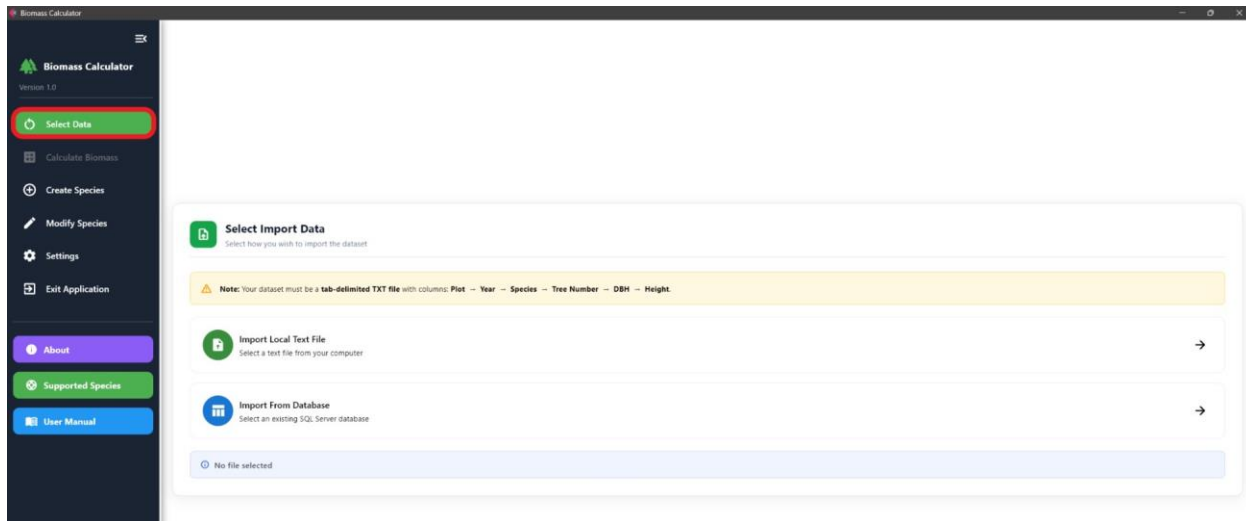


Figure 3. The [Select Data] button is highlighted.

4.a - Importing from Text File

To import a text file, select [Import Local Text File], which will prompt you to select a text file to use.

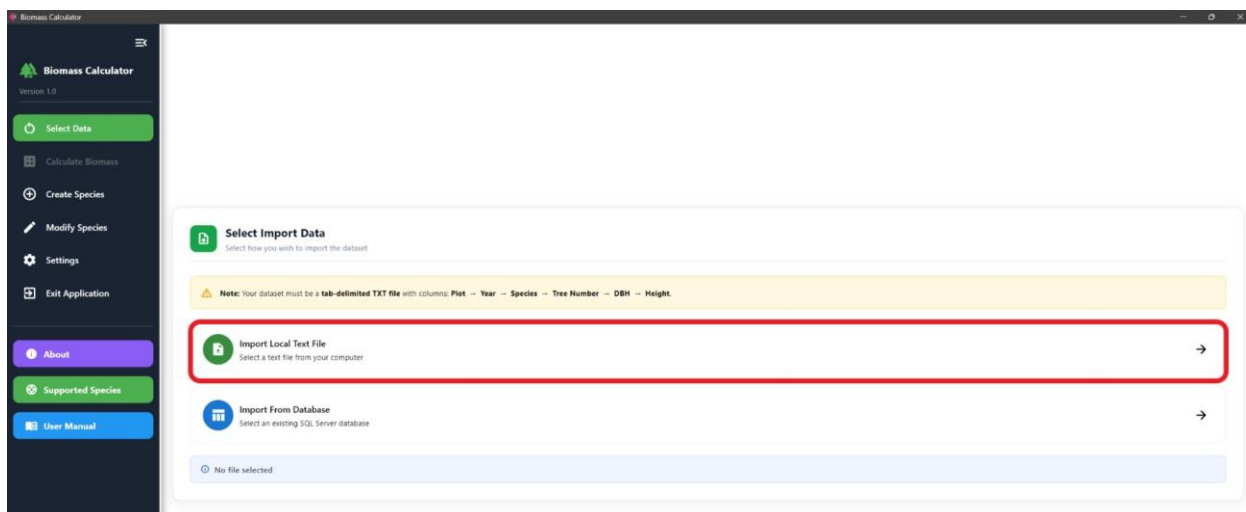


Figure 4. The [Import Local Text File] button is highlighted.

The dataset columns in the text file must have the core fields (headers): **Plot, Year, Species, Tree Number, DBH, Height**.

Tip: The text dataset should be made using Excel or exported from the database, so tabs and columns are properly aligned.

For example:

Plot	SubPlot	Year	Origin	Tree	Status	Species	Tree Number	DBH	Height
1108656	1	2005	Natural	Stand	L	251	2	18.3	14.48

Figure 5. Screenshot of the fields in a text file.

If done correctly, it should show the name of the file selected at the bottom.

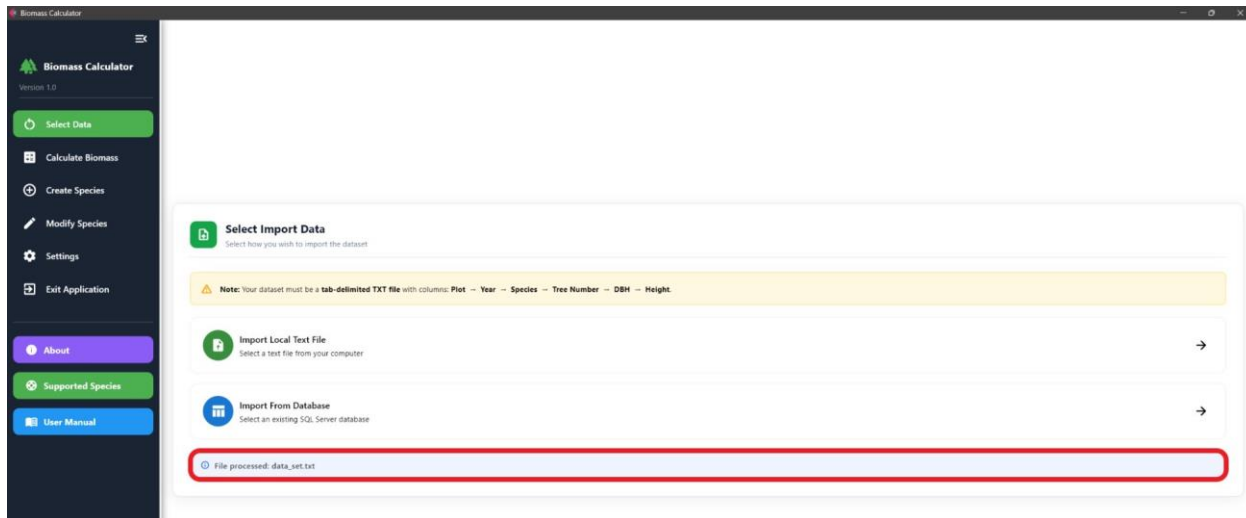


Figure 6. The confirmation of the file processed is highlighted.

4.b - Importing from Database

To **import from a database**, select “**Import From Database**”, which will **open a dialog** to enter details to **connect to a database**. To select this option, users’ machines need to host a SQL server, so the tool can connect to the database directly. Preferably, users should have **Microsoft SQL Express** installed as the **SQL server's engine**, and **Microsoft SQL Server Management Studio** as the graphical user interface to interact with the SQL server.

- **SQL Server 2025 Express:** <https://www.microsoft.com/en-us/sql-server/sql-server-downloads>
- **Microsoft SQL Server Management Studio:** <https://learn.microsoft.com/en-us/ssms/install/install>

Once host machines have both software installed and executed, the tool will automatically detect the SQL server's engine. Users just need to type in the name of the database needed for biomass calculation, and the tool will open a connection to that database to get input data from and write output data.

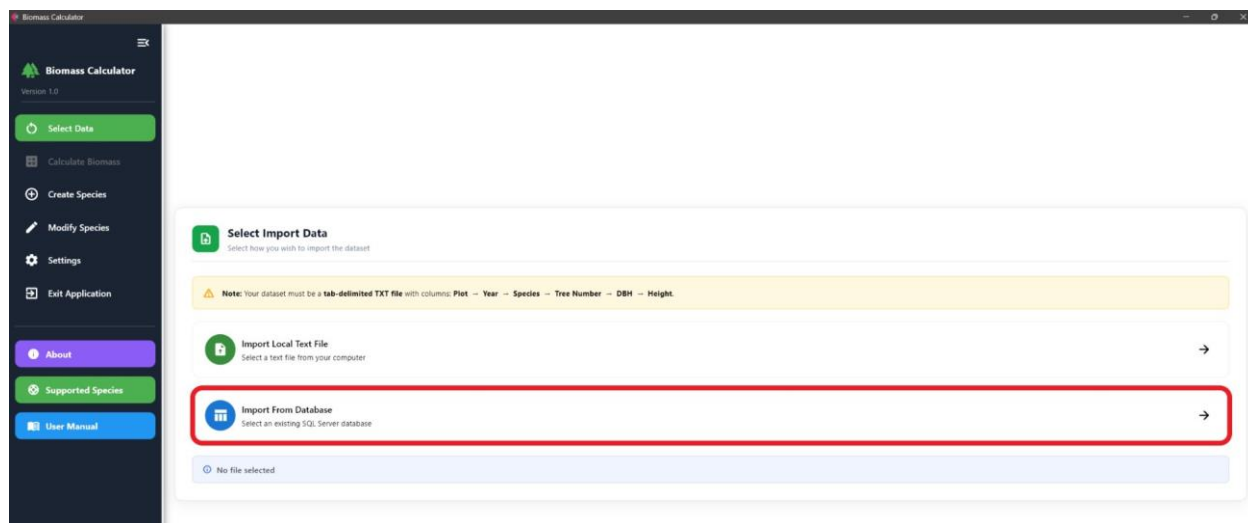


Figure 7. The [Import From Database] button is highlighted.

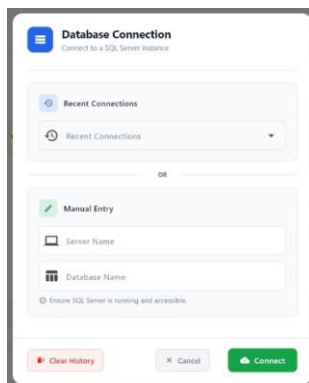


Figure 8. The opened menu for selecting a database (no SQL server detected so all fields are empty)

To **select a database**, you can **either select a recently connected database** by opening the drop-down menu and clicking on one of the connections, or you can **manually enter a server name and database name** into the corresponding text boxes to connect to (in case the tool cannot detect any running SQL server).

Note: In case users connect to “gyPSPPGP” database, the input data comes from “dbo.tCalcBiomassInput” table. The table for storing calculated biomass data is named “dbo.tCalcBiomassOutputD” or “dbo.tCalcBiomassOutputDH”.

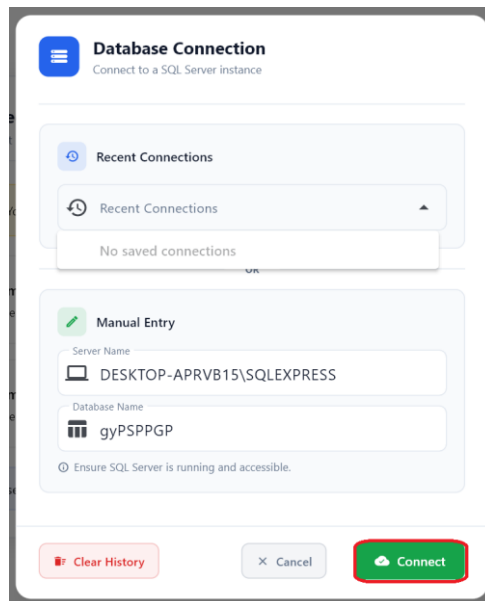


Figure 9. Manually enter “**Server Name**” and “**Database Name**”, and then click [Connect] to open connection for the first time (no connection history)

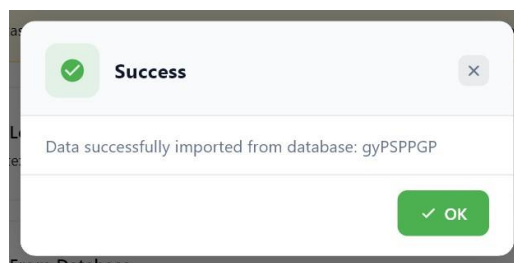


Figure 10. Notice when the tool successfully connects to a database and imports data from it

Note: For the dataset import from the database to work, it needs to have a table called **"tCalcBiomassInput"** with **Plot, Year, Species, Tree Number, DBH, Height** columns

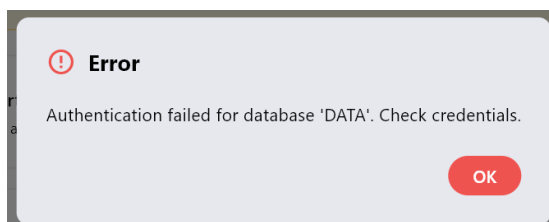
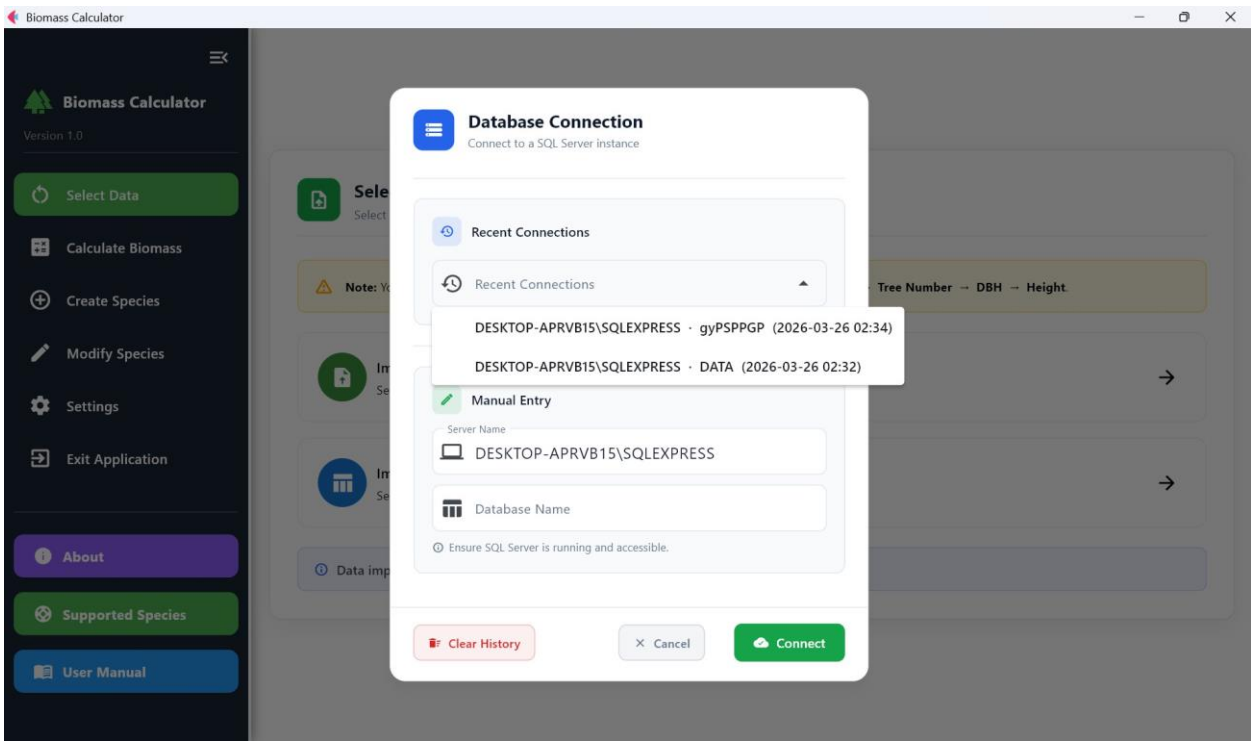


Figure 11. Notice when user enters incorrect server name and/or database name, or doesn't have permission to access such database

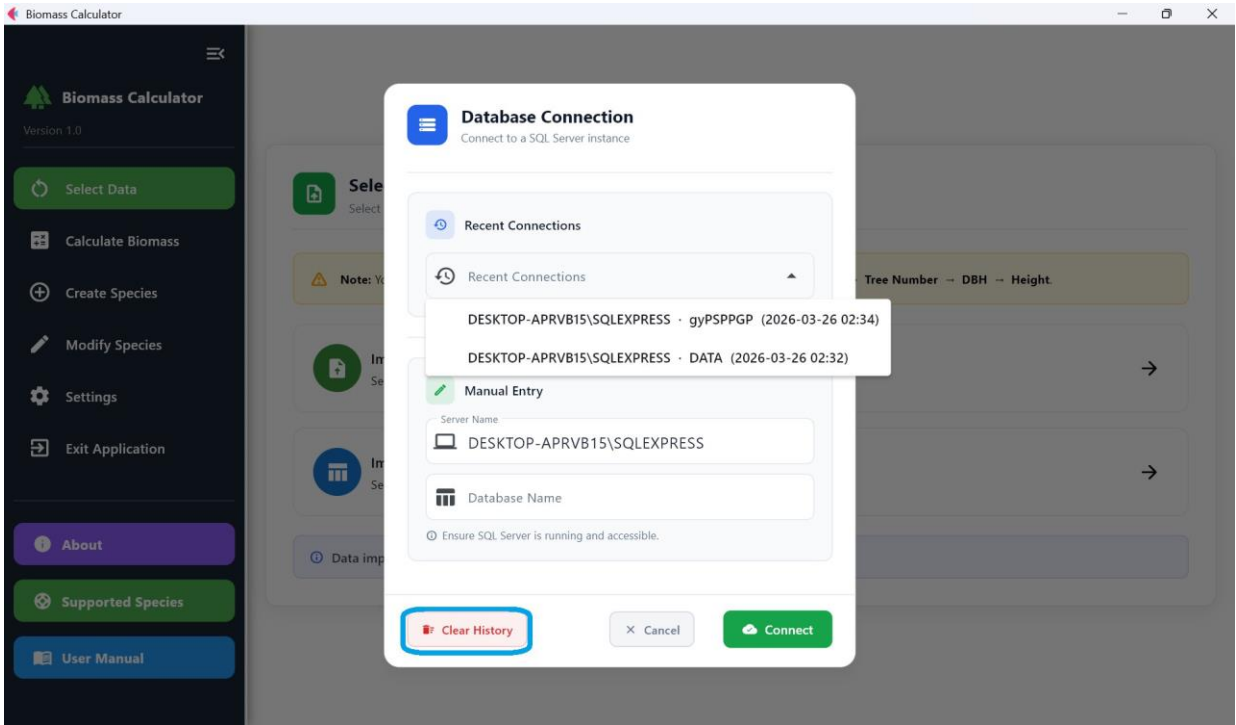
If the tool has made connections in the past, clicking on the drop-down arrow in **"Recent Connections"** will show all connection logs. Clicking on any will auto-fill the fields **"Server Name"** and **"Database Name"**. Then click on the [Connect] button to confirm.



Figure

Figure 12. “Recent Connections” shows logs

To clear the history of all recent connections, click on the red “Clear History” button. After performing that, clicking on the drop-down arrow will show no logs.



Figure

Figure 13. Before clearing “Recent Connections”. The [Clear History] button is highlighted.

5) Calculate Biomass

After selecting input data, click on the [Calculate Biomass] button on the left-hand side menu to specify which type of computation to perform on the data.

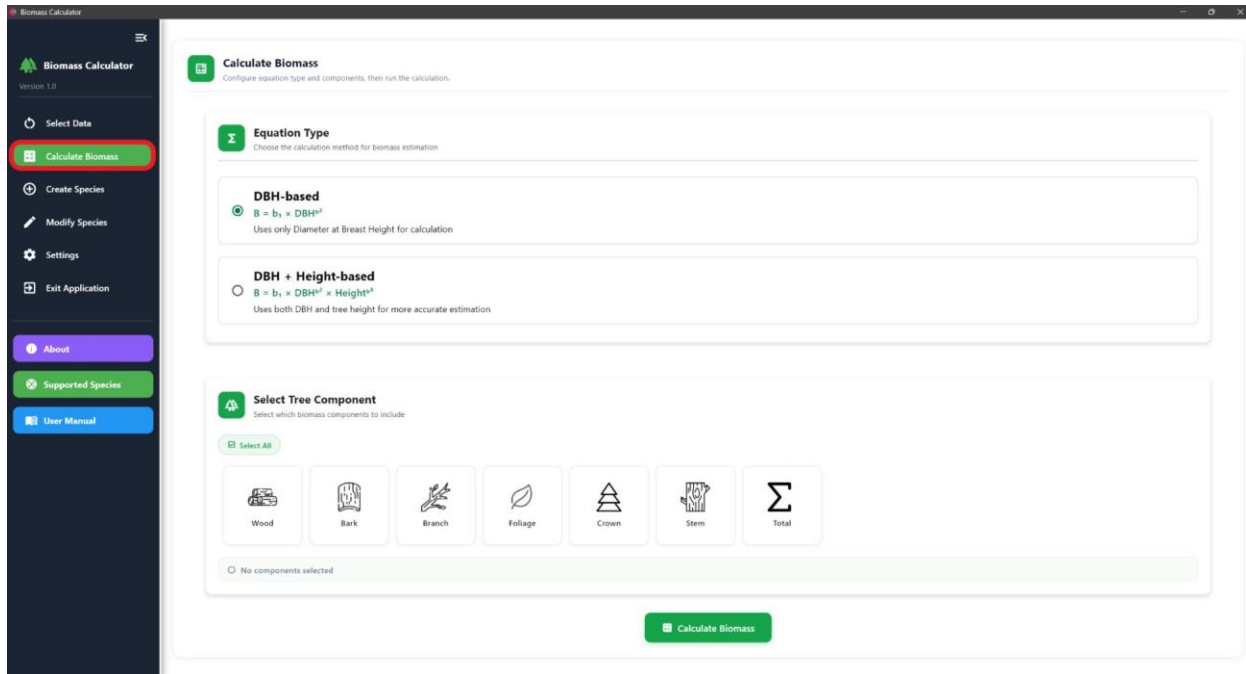


Figure 14. App screen after clicking on [Calculate Biomass]

5.a - Selecting Equation + Components

In the “**Equation Type**” section of the [Calculate Biomass] screen, users can **choose one of the two biomass-calculating functions** simply by clicking on the one desired: **DBH-based (simple)** or **DBH-and-height-based (more accurate)**. The default selection is DBH-based.

Also, users can select to have all biomass features to be accounted for in the equation or select some features and omit the rest for calculation. A feature when selected (by clicking on it) has the green background and outline, and clicking on that feature again deselects it (and its background turns white).

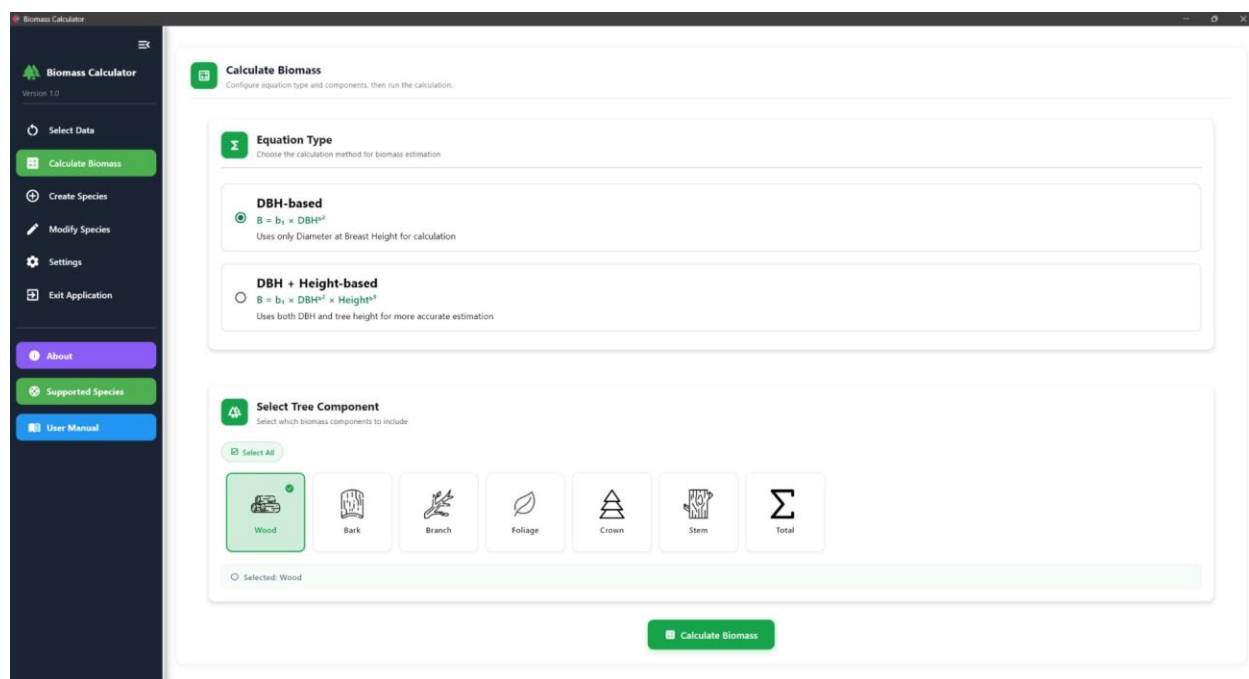


Figure 15. Only “Wood” feature being selected

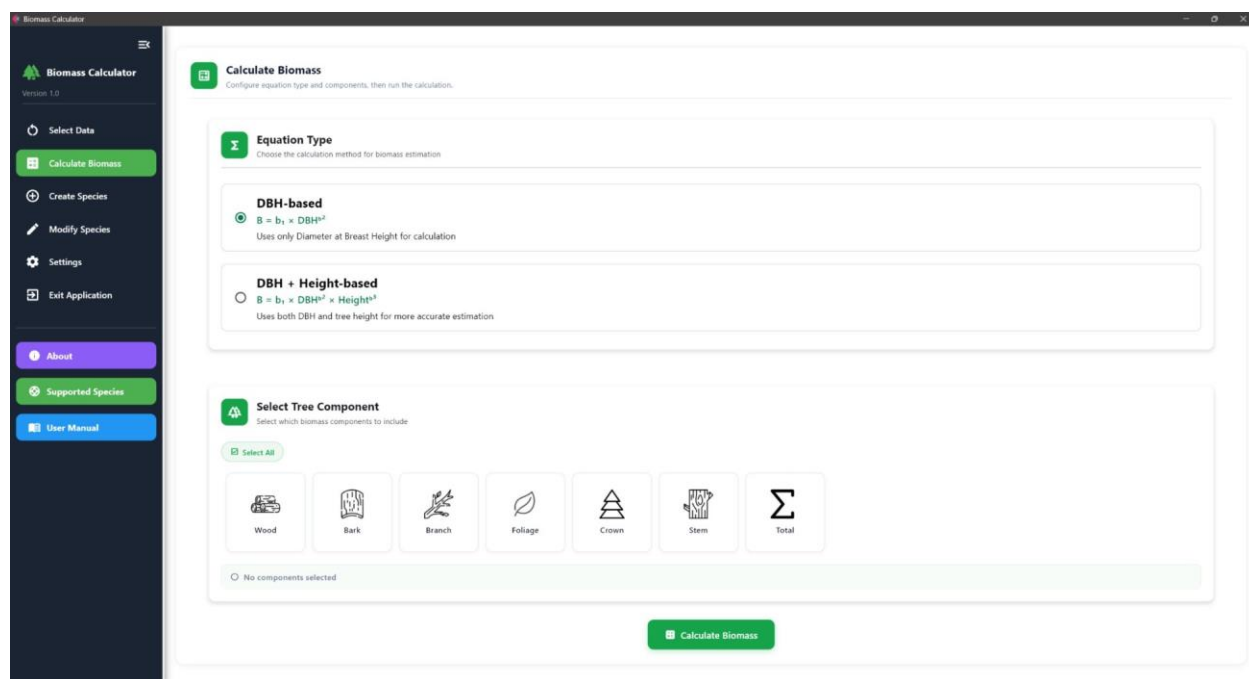


Figure 16. Clicking on “Wood” feature again to deselect it (No feature is selected)

To **quickly select all features**, click on the **“Select all”** check button. Once all features are selected, the “Select all” check button will change to the “Deselect all” check button that allows users to remove all features from selection.

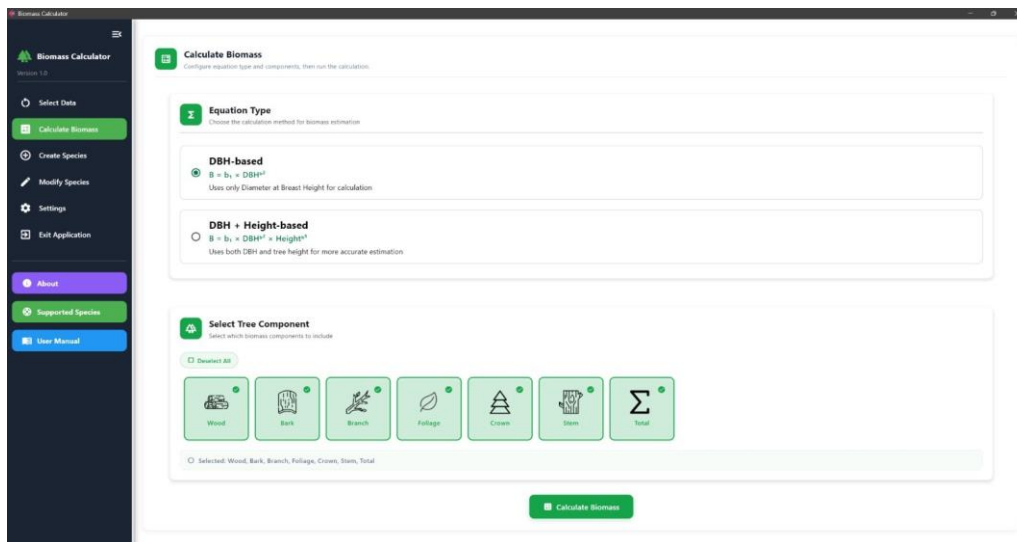


Figure 17. After clicking “Select All” button, the button will change to “Deselect all”

Note: If users import data from a database for calculation, **all tree components will automatically be selected** and cannot be chosen to be omitted. In this mode, **users can only select DBH-based equation or DBH-and-Height-based equation**.

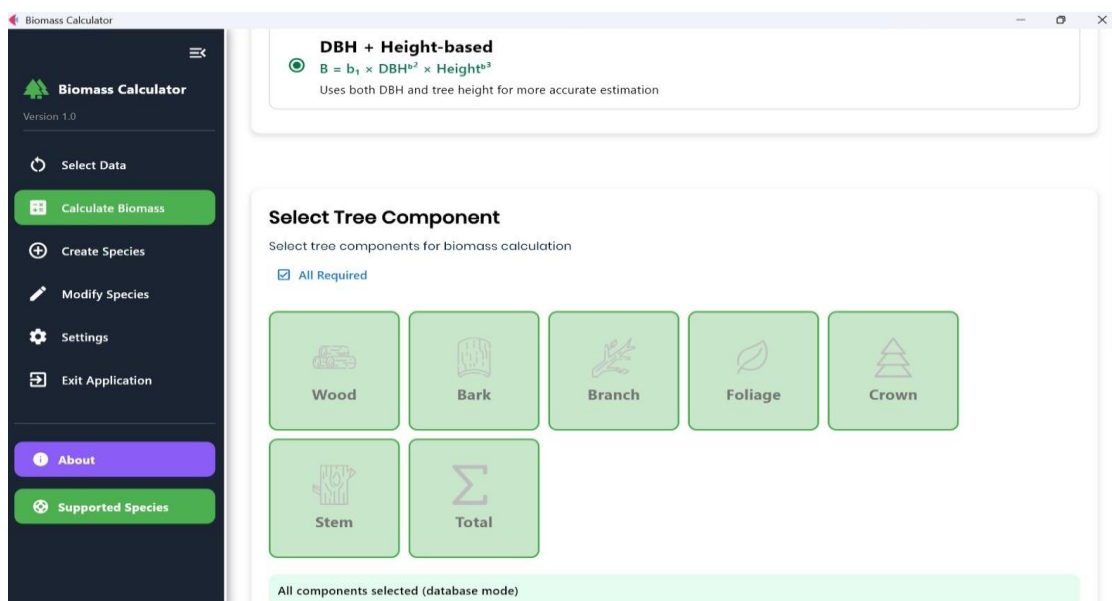


Figure 18. All features must be selected (automatically by default) in database mode

5.b - Species Classification (Only in Text-file Mode)

After selecting the type of function and features for calculation, users can click on the **“Calculate Biomass”** button at the bottom of the screen. At this point, a **pop-up will show up asking users to define the codes for hardwood and softwood prior to if the tool does not support that species or user has not manually created it.**

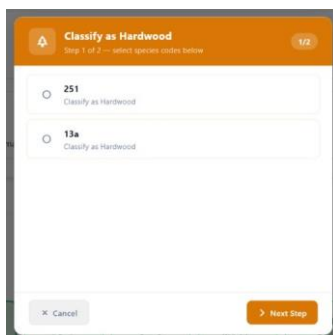


Figure 19. Pop-up after clicking “Calculate Biomass”

After selecting the type of function and features for calculation, users can click on the **“Calculate Biomass”** button at the bottom of the screen. At this point, a pop-up will show up asking users to define the codes for hardwood and softwood. Users will be asked to choose one of the two (or more) options as the codename of Hardwood and can proceed when clicking on “Next Step.” The leftover options after this step will be the available option(s) as the codename for Softwood. Choose the codename for Softwood by clicking on one, and then users can click on the “Submit Final” button to start the calculation.

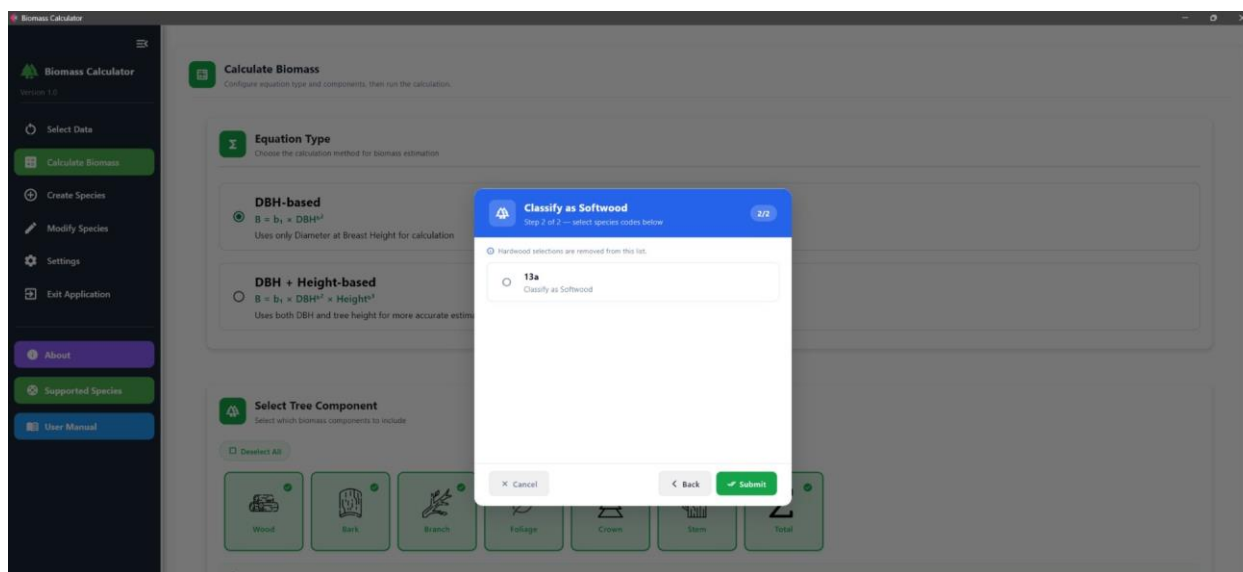


Figure 20. The leftover option(s) after selecting codename for Hardwood. Clicking on “Submit Final” button after choosing a codename for Softwood to start calculating.

5.c - Exporting Results

After the calculation is done, users can preview the first 10 records of the returned results. Horizontal scroll on the result preview panel is enabled to allow users to see all columns of the generated dataset(s).

The screenshot shows the Biomass Calculator interface. The top section allows users to select biomass components to include: Wood, Bark, Branch, Foliage, Crown, Stem, and Total. Below this is a 'Calculate Biomass' button. The main panel displays 'Calculated Biomass Results' with a table showing the first 10 entries of the generated dataset. The table has 12 columns: plot, subplot, year, origin, tree status, species, tree number, dbh, height, Wood (kg), Bark (kg), Foliage (kg), Branch (kg), Crown (kg), Stem (kg), and Total (kg). The data is as follows:

plot	subplot	year	origin	tree status	species	tree number	dbh	height	Wood (kg)	Bark (kg)	Foliage (kg)	Branch (kg)	Crown (kg)	Stem (kg)	Total (kg)
1108656	1	2005a	Natural Stand	L	13a	1a	19.5a	14.07	N/A	N/A	N/A	N/A	N/A	N/A	N/A
1108656	1	2005	Natural Stand	L	251	2	18.3	14.48	85.6	14.3	4.4	22.6	27.0	99.9	126.9
1108656	1	2005	Natural Stand	L	251	3	17	13.84	71.9	12.2	3.9	18.8	22.7	84.0	106.7
1108656	1	2005	Natural Stand	L	13	4	17.8	13.82	66.5	9.8	9.7	11.2	20.9	76.3	97.2
1108656	1	2005	Natural Stand	L	13	5	15.7	13.81	48.5	7.4	8.1	8.6	16.7	55.9	72.6
1108656	1	2005	Natural Stand	L	13	6	18.2	13.64	70.3	10.3	10.0	11.7	21.7	80.6	102.3
1108656	1	2005	Natural Stand	L	13	7	15.7	13.32	48.5	7.4	8.1	8.6	16.7	55.9	72.6
1108656	1	2005	Natural Stand	L	13	8	16	12.08	50.9	7.7	8.3	9.0	17.3	58.6	75.9
1108656	1	2005	Natural Stand	L	13	9	10.3	10.77	16.8	2.9	4.5	3.6	8.0	19.7	27.7
1108656	1	2016	Natural Stand	L	13	95	10.3	10.77	16.8	2.9	4.5	3.6	8.0	19.7	27.7

The interface also includes a sidebar with navigation options: Select Data, Calculate Biomass, Create Species, Modify Species, Settings, Exit Application, About, Supported Species, and User Manual. The bottom of the table shows 'Page 1 of 2' and navigation buttons for 'Previous', '1', '2', and 'Next'.

Figure 21. The result preview panel showing the first 10 entries of the generated dataset. Users can scroll horizontally to see all the columns

5.c [i] - Exporting to Text File

If the input data is from a text file, then users can only export the output data to a new text file. To write the calculated data to a new text file, click on the “Export to TXT” button to open up a file-saving dialogue.

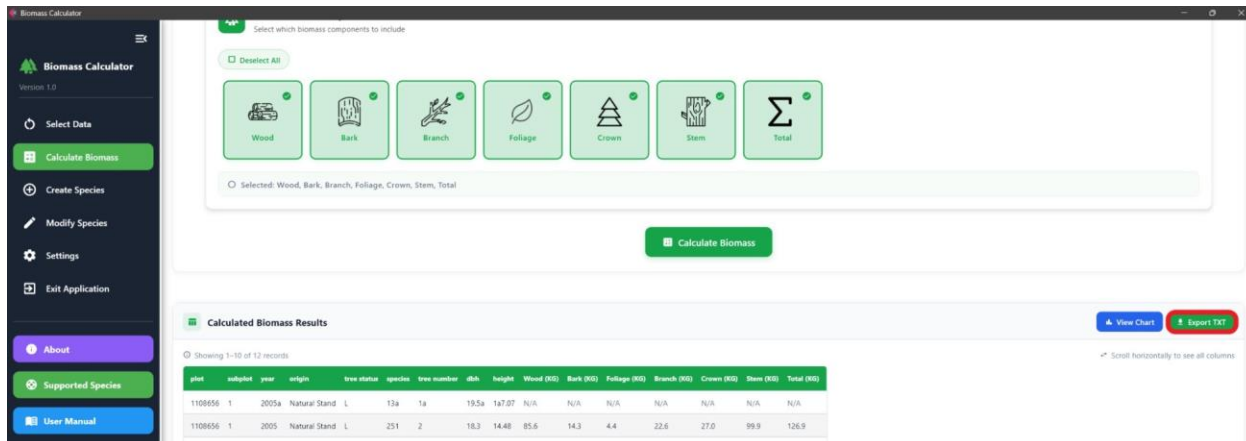


Figure 22. Click on [Export to TXT] button to open file-saving dialogue

5.c [ii] - Writing to Database

This option is only available if the input data comes from a database. Clicking on the orange [Write to Database] button will save output data to the same database of input data.

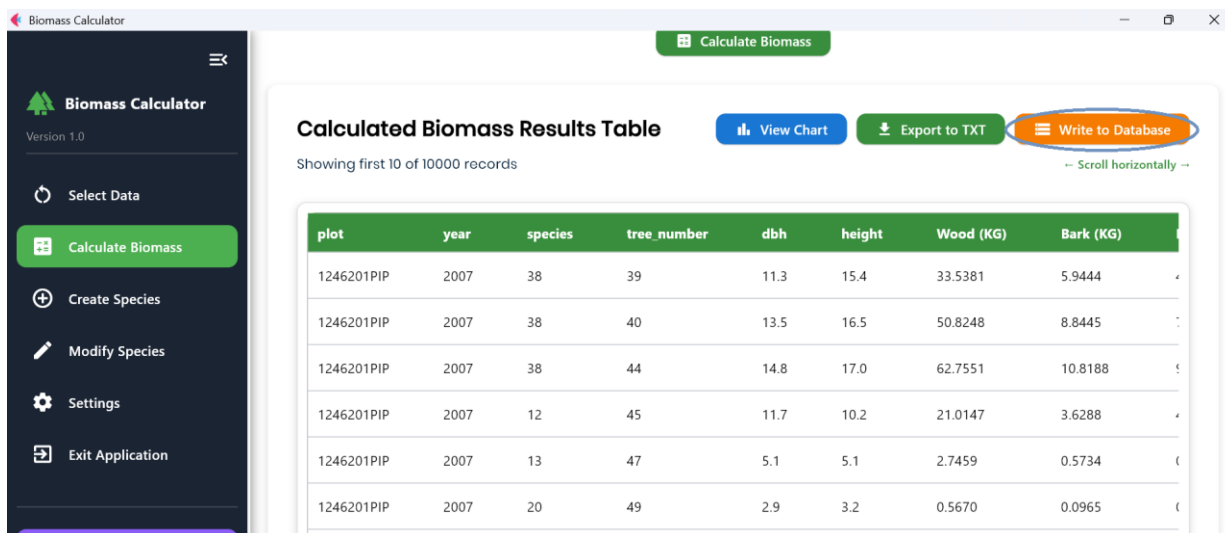


Figure 23. Preview panel after the biomass calculation is done. Choose [Write to Database] to save data to preconfigured table

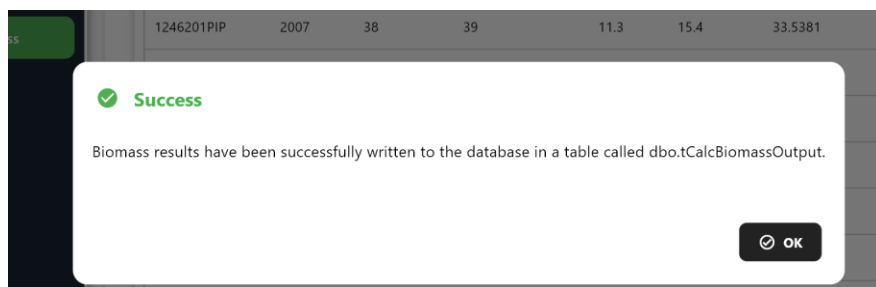


Figure 24. Notice when output data is successfully recorded in database

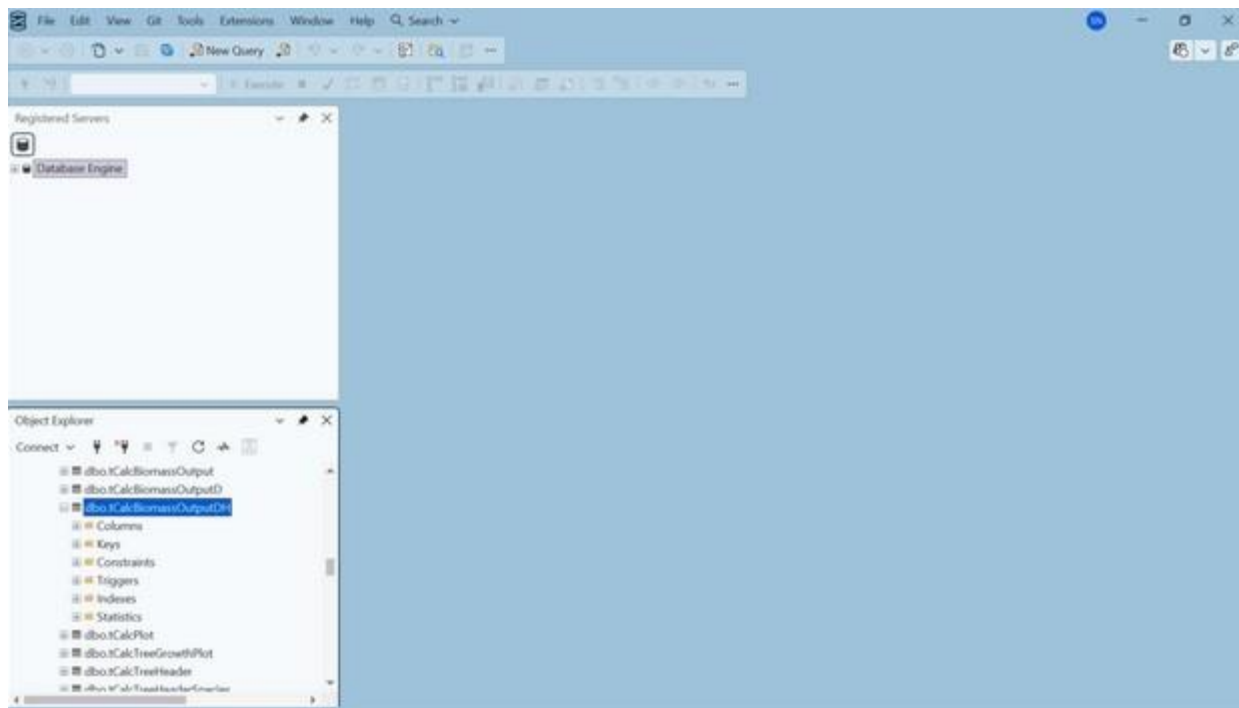


Figure 25. For DBH equation, output is saved to dbo.tCalcBiomassOutputD and for DBH + Height equation, output is saved to dbo.tCalcBiomassOutputDH

5.c [iii] - Generating Plot

Users can see plot(s) mapped from data points in the calculated output. To **view the graph(s)**, click on the blue **[View Chart]** button, and the graph(s) can be saved as individual files.

The screenshot shows the 'Biomass Calculator' application interface. On the left is a sidebar with navigation options: Select Data, Calculate Biomass, Create Species, Modify Species, Settings, Exit Application, About, Supported Species, and User Manual. The main area is titled 'Select which biomass components to include' and shows a selection of components: Wood, Bark, Branch, Foliage, Crown, Stem, and Total. Below this is a 'Calculate Biomass' button. The 'Calculated Biomass Results' panel is displayed, showing a table of results. The table has columns for plot, subplot, year, origin, tree status, species, tree number, dbh, height, Wood (kg), Bark (kg), Foliage (kg), Branch (kg), Crown (kg), Stem (kg), and Total (kg). The table shows 12 records, with the first 10 displayed. A 'View Chart' button is located in the top right corner of the results panel.

plot	subplot	year	origin	tree status	species	tree number	dbh	height	Wood (kg)	Bark (kg)	Foliage (kg)	Branch (kg)	Crown (kg)	Stem (kg)	Total (kg)
1108056	1	2005a	Natural Stand	L	13a	1a	19.5a	147.07	N/A	N/A	N/A	N/A	N/A	N/A	N/A
1108056	1	2005	Natural Stand	L	251	2	18.3	14.48	85.6	14.3	4.4	22.6	27.0	99.9	126.9
1108056	1	2005	Natural Stand	L	251	3	17	13.84	71.9	12.2	3.9	18.8	22.7	84.0	106.7
1108056	1	2005	Natural Stand	L	13	4	17.8	13.82	66.5	9.8	9.7	11.2	20.9	76.3	97.2
1108056	1	2005	Natural Stand	L	13	5	15.7	13.81	48.5	7.4	8.1	8.6	16.7	55.9	72.6
1108056	1	2005	Natural Stand	L	13	6	18.2	13.64	70.3	10.3	10.0	11.7	21.7	80.6	102.3
1108056	1	2005	Natural Stand	L	13	7	15.7	13.32	48.5	7.4	8.1	8.6	16.7	55.9	72.6
1108056	1	2005	Natural Stand	L	13	8	16	12.08	30.9	7.7	8.3	9.0	17.3	58.6	75.9
1108056	1	2005	Natural Stand	L	13	9	10.3	10.77	16.8	2.9	4.5	3.6	8.0	19.7	27.7
1108056	1	2016	Natural Stand	L	13	95	10.3	10.77	16.8	2.9	4.5	3.6	8.0	19.7	27.7

Figure 26. Click [View Chart] button to see generated graph(s) from data points from output

Users can see plot(s) mapped from data points in the calculated output. To view the graph(s), click on the blue [View Chart] button, and the graph(s) can be saved as individual files. The Download icon on the top right corner of the chart panel allows users to save the graph(s) as either a png or a csv file. To quit that panel, click on the [X] button to the right of the Export icons to go back to the output preview panel.

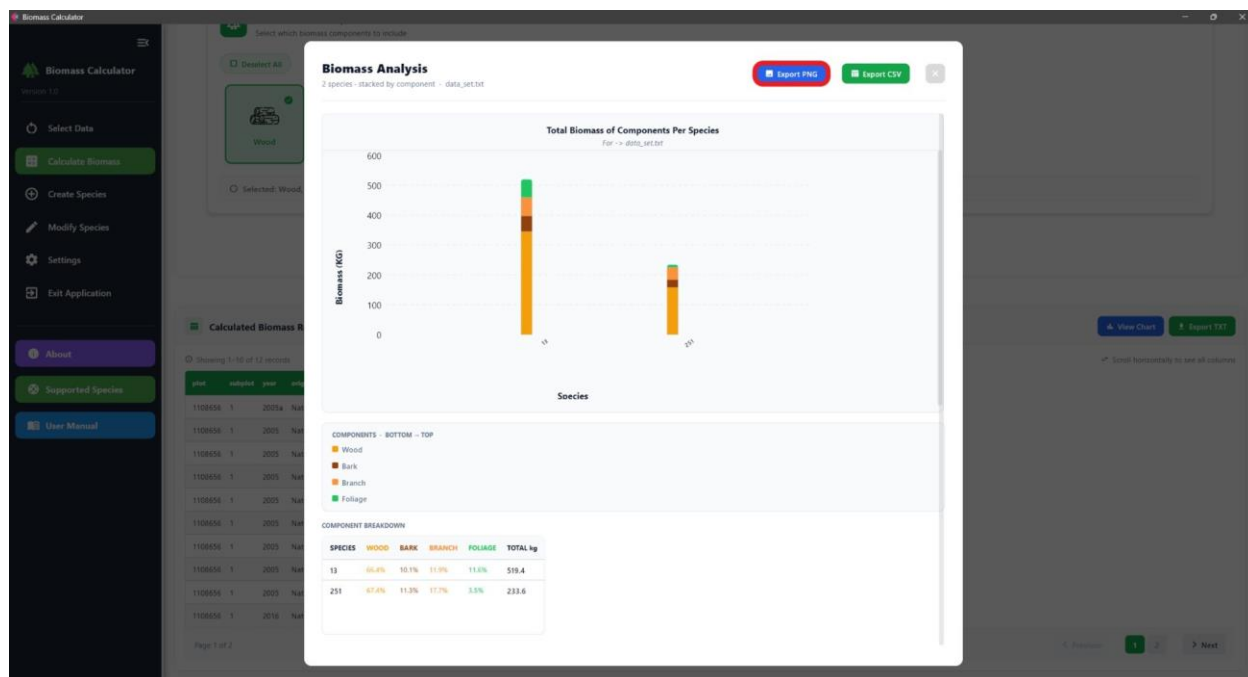


Figure 27. Click Export png icon (red highlight) to save chart as an individual png

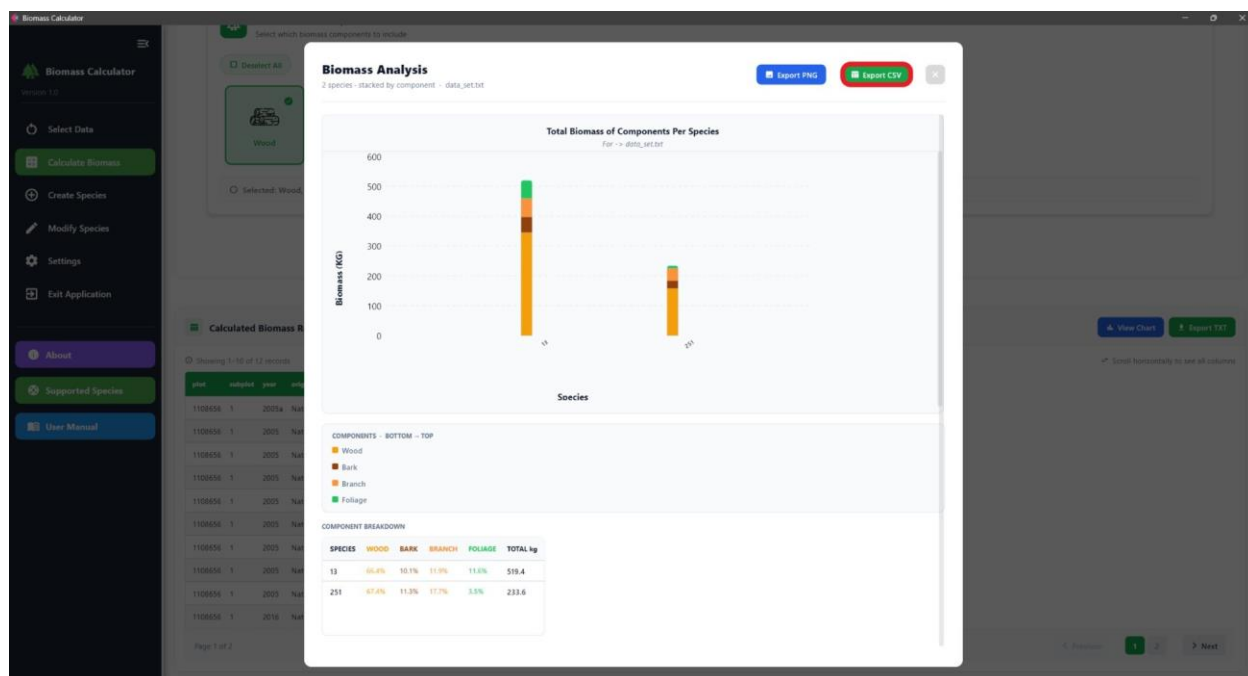


Figure 28. Click Export csv icon (red highlight) to save chart as a csv file

6 - Create Species

To go to the “Create Species” console, click on the [Create Species] button on the left-hand menu, and users will be shown the “Create Species” screen.

Figure 29. Click [Create Species] button (red outline) to show the console

Users will be asked to put in mandatory details of the new tree species (name or code number, origin, applicable equation type, tree features to be accounted for in the biomass function, and parameter values corresponding to select features for calculation).

In the “Parameters” section, all values to fill must be in the (-5.0, 5.0) range. If users omit or enter invalid values, warning messages in red will appear at the bottom of the respective fields.

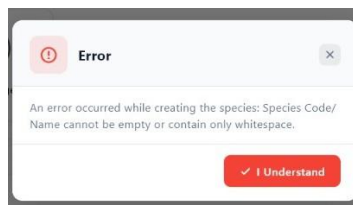


Figure 30. Warning messages appear when fields left empty or filled with invalid values

If all necessary fields are filled with acceptable values, a preview panel will pop up to allow users to see all details of the newly entered species to make sure everything is correct. Click

on the [Proceed] button at the bottom-right corner to save the new tree species or [Cancel] to go back to the “Create Species” console.

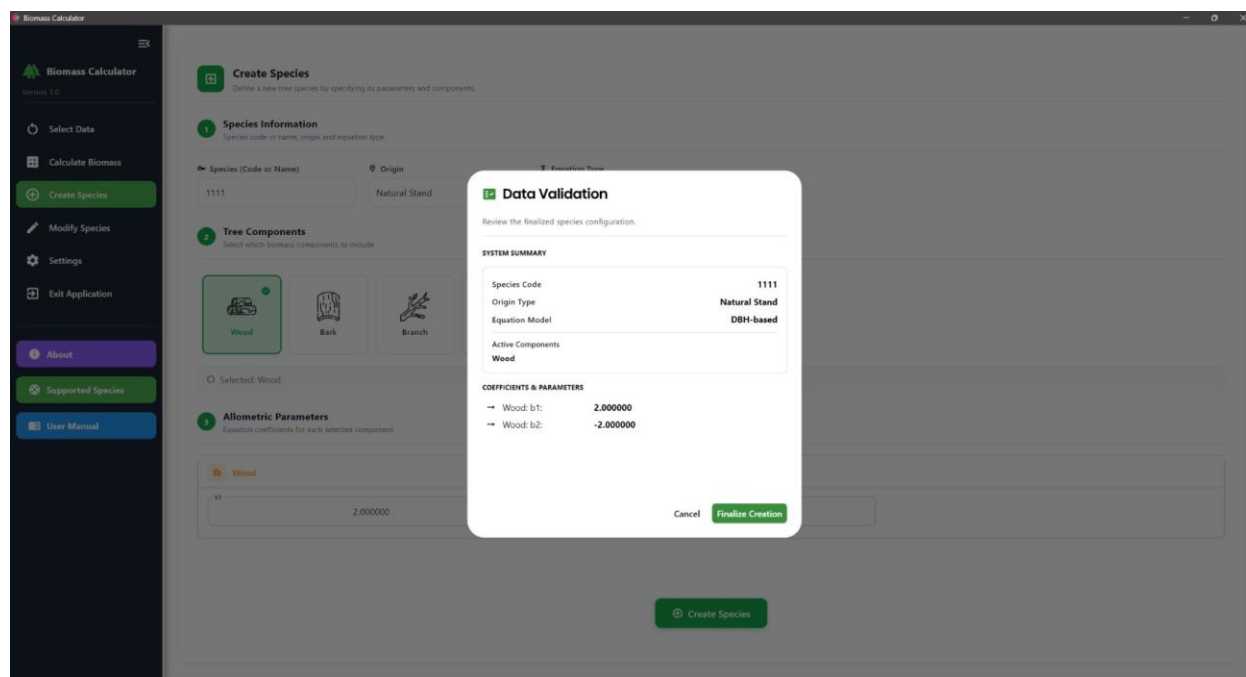


Figure 31. Preview panel before saving a new tree species.

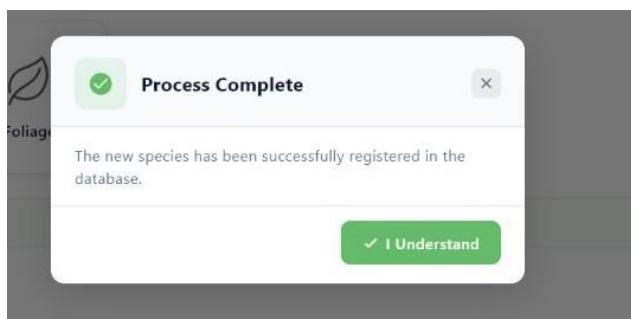


Figure 32. Notice when new tree species added to the tool’s memory successfully

7) Modify Species

To modify a created species, open the **“Modify Species”** menu by clicking on the Modify Species button on the left sidebar.

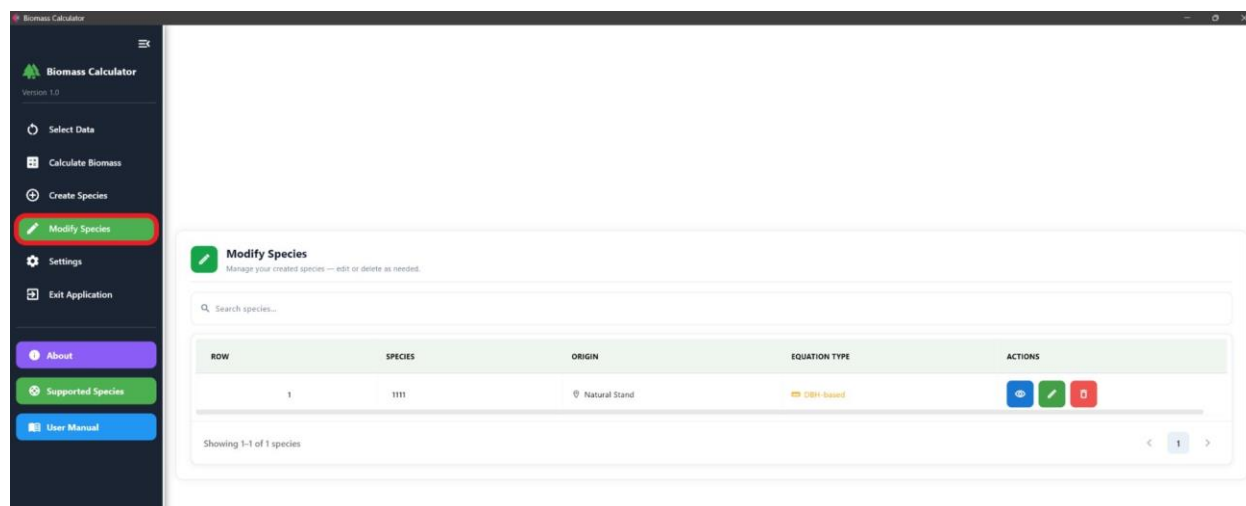


Figure 33. The [Modify Species] button is highlighted.

7.a - Search for Species

To search for a specific species, enter the code, common name, or origin of the species into the search bar.

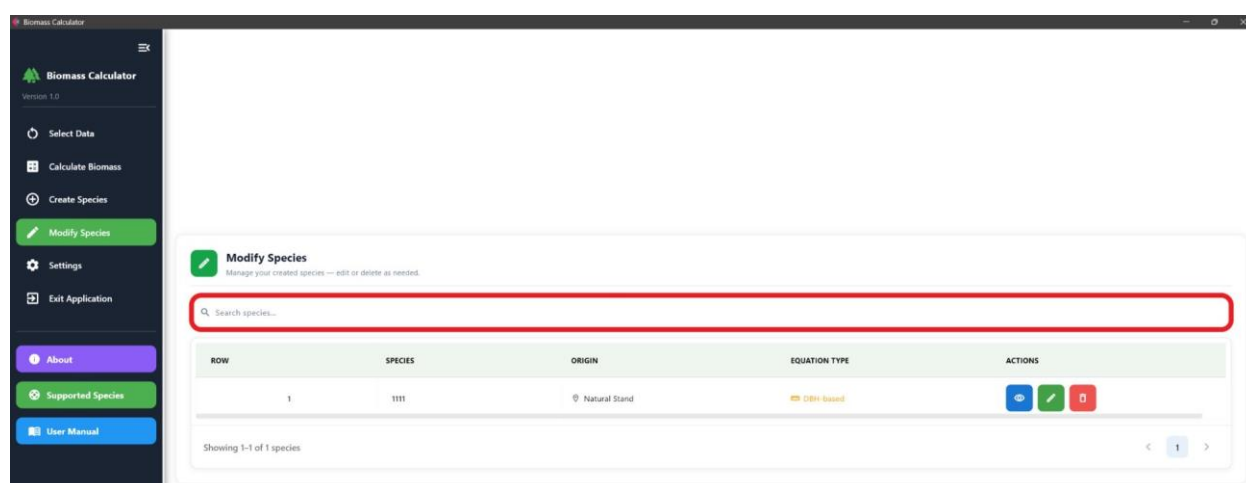


Figure 34. The Modify Species search bar is highlighted.

7.b - Species Details

To view specific species details, click on the blue Species Details button indicated by an eyeball.

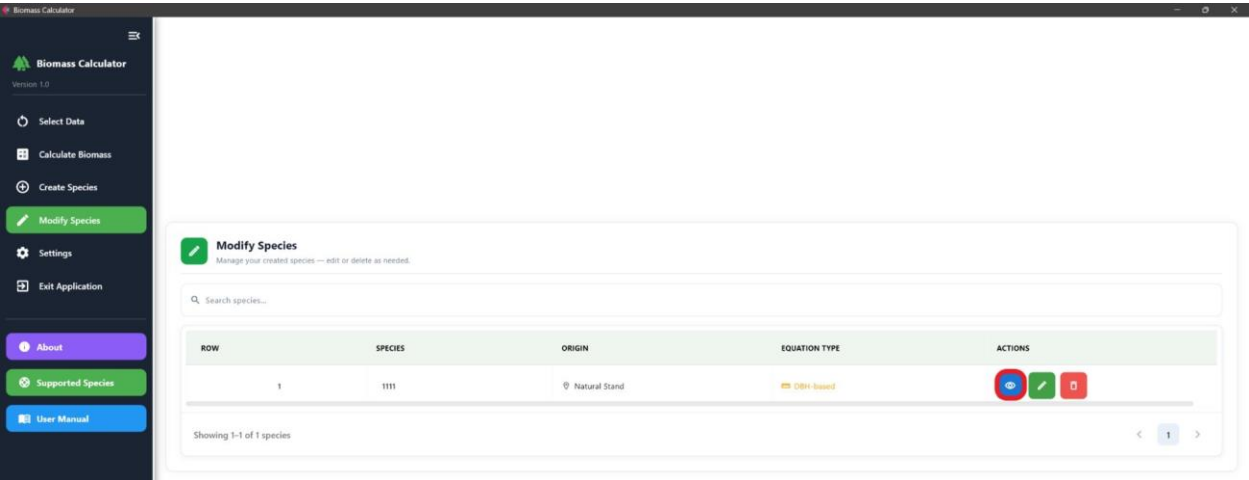


Figure 35. The [Species Details] buttons are highlighted.

You can scroll through the menu that pops up to view all the details for that species.

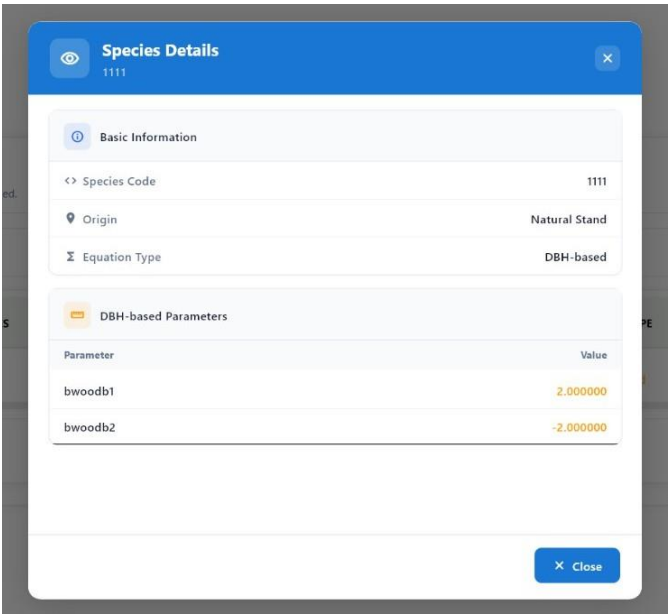


Figure 36. The Species Details menu.

7.c - Edit Species

To edit a specific species, click on the green Edit Species button indicated by a pencil.

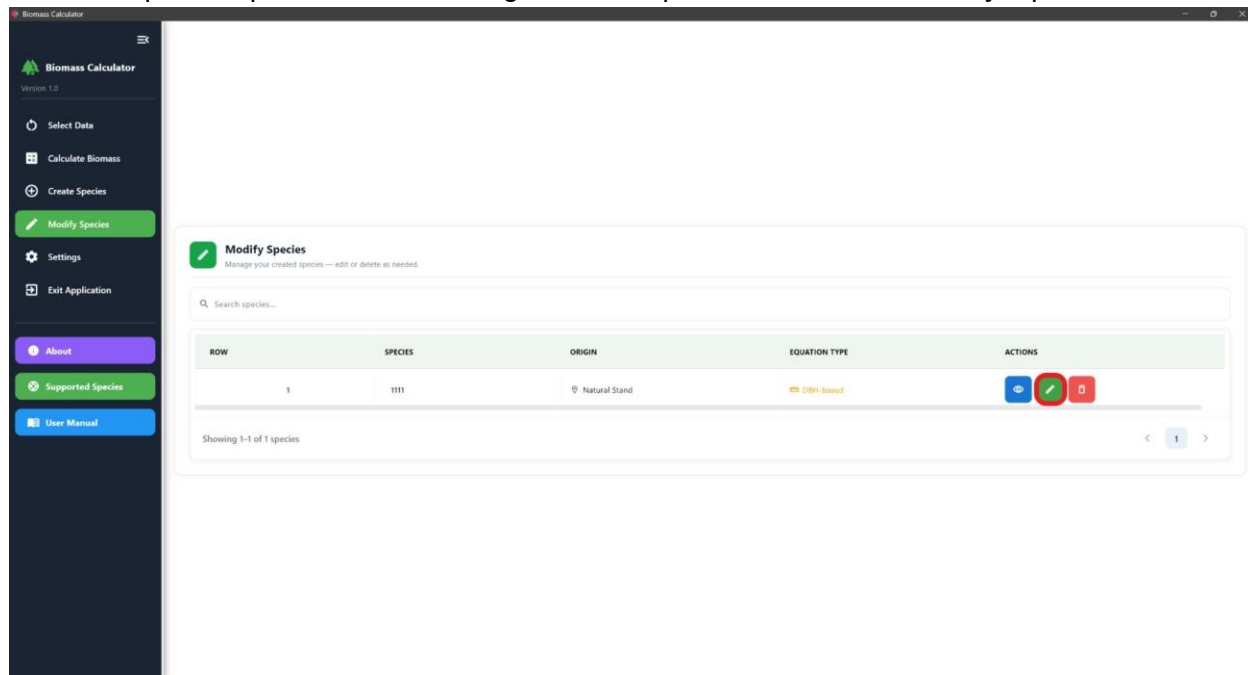


Figure 37. The Edit Species buttons are highlighted.

You can scroll through the menu that pops up and edit any of the details of that species.

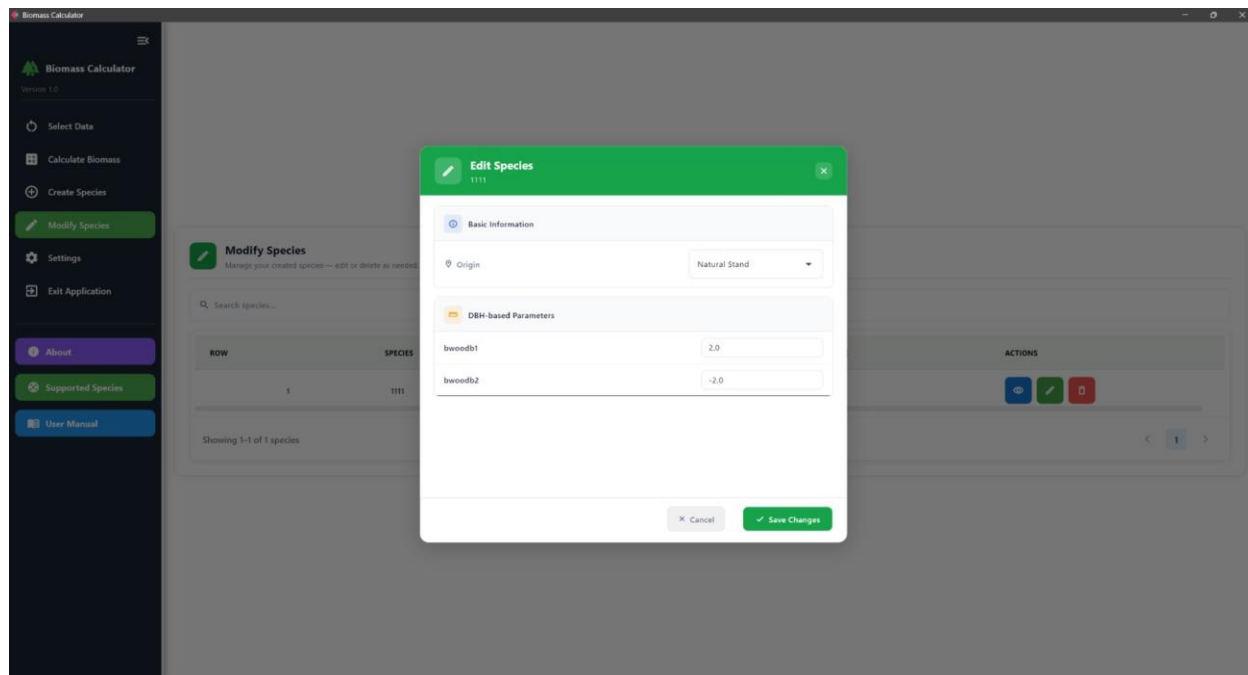


Figure 38. The Edit Species menu.

7.d - Delete Species

To delete a specific species, click on the red Delete Species button indicated by a garbage can.

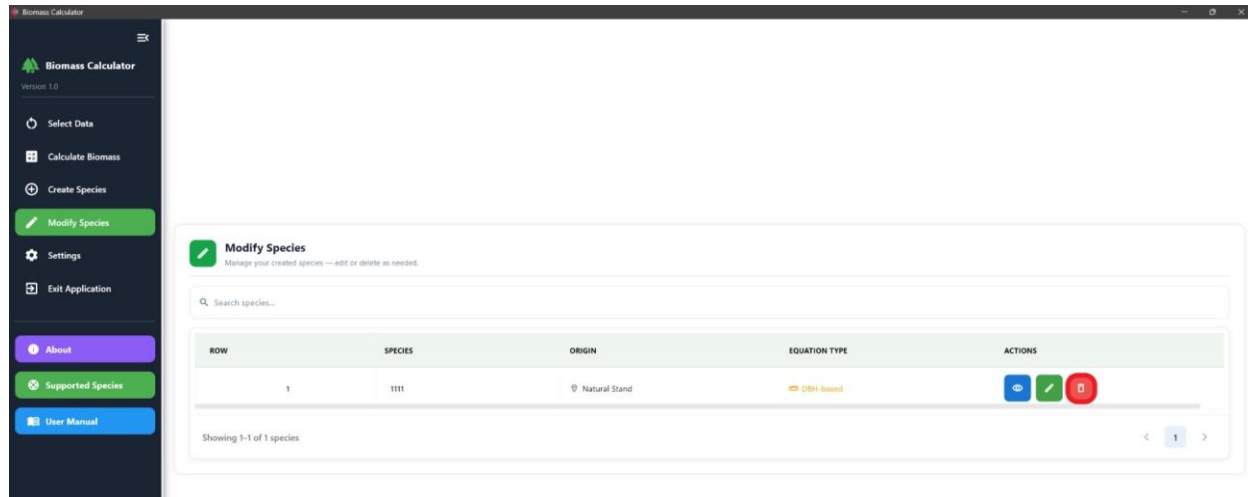


Figure 39. The Delete Species buttons are highlighted.

You then must confirm the delete by pressing the “Delete Species” button on the bottom of the menu.

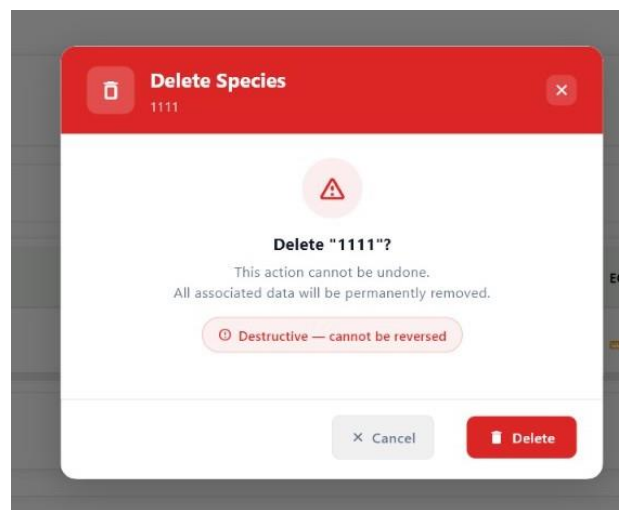


Figure 40. The Confirm Delete menu.

The most recent version of the Modify Species page also has Load **Species** and **Save Species** and **Clear Data** buttons.



Figure 41. The Load Species, Save Species and Clear Data buttons

7.e – Clear Data

Clear Data makes it so that all the user created species within the table will be permanently deleted.

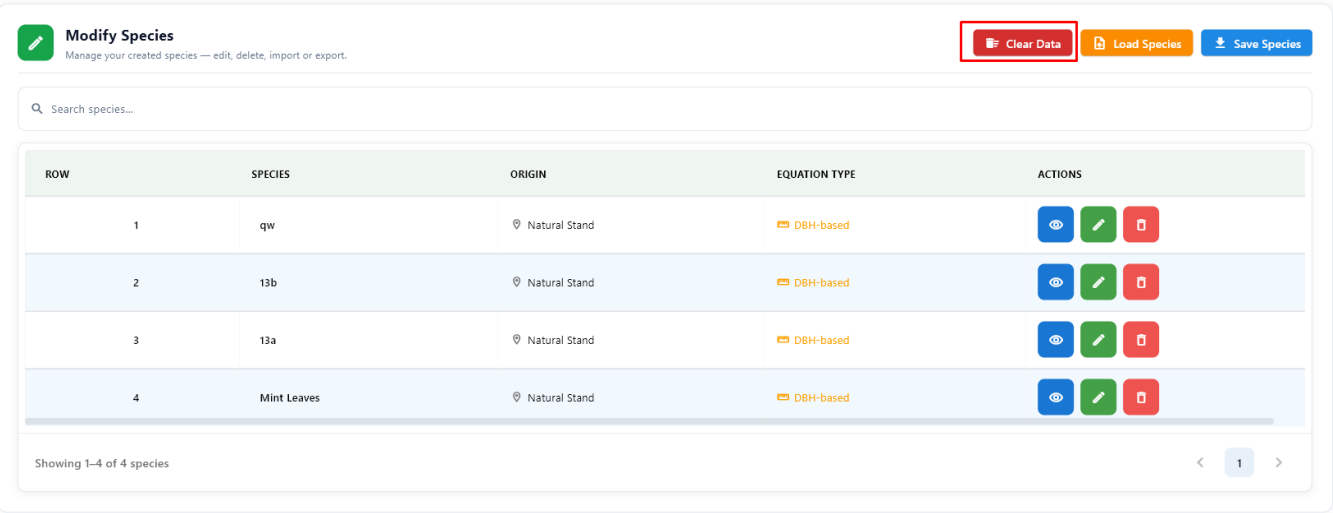


Figure 42. Clear Data button is clicked

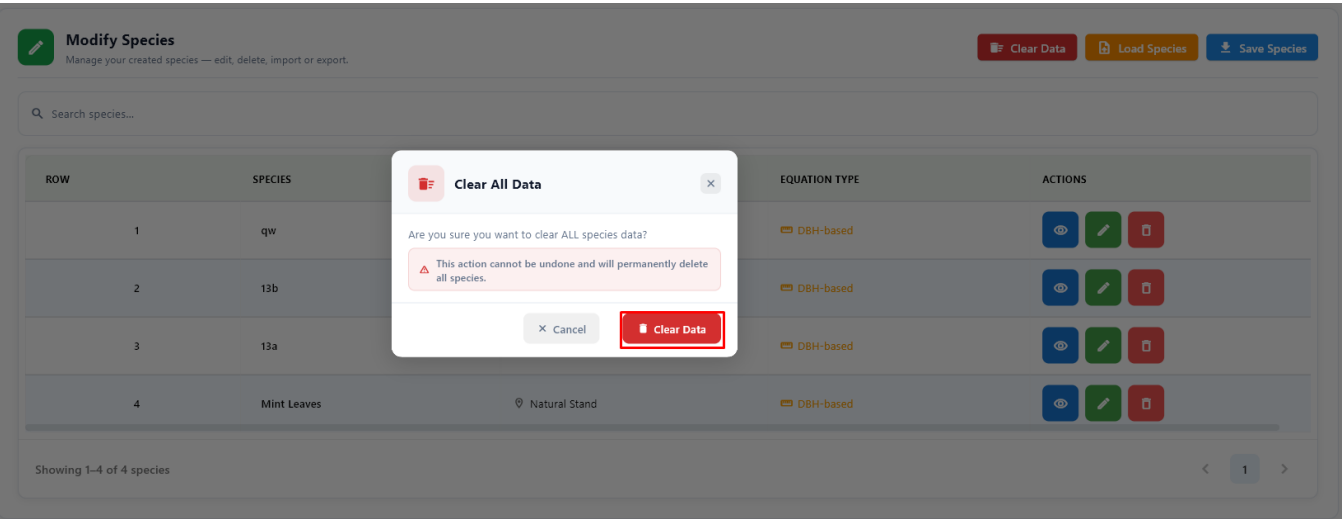


Figure 43. Confirmation dialog of deletion

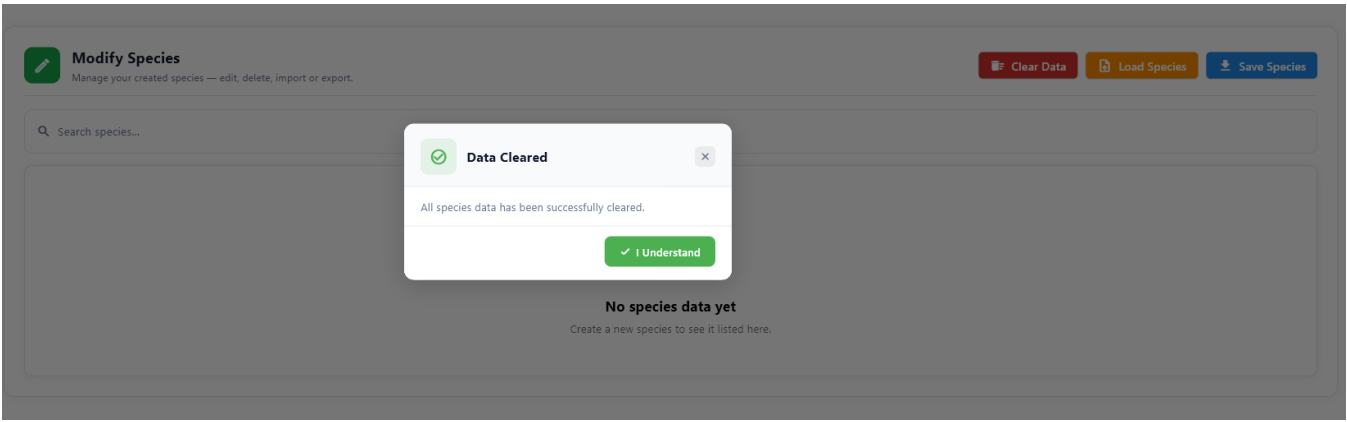


Figure 44. Successful Deletion of user created species

7.f – Load Species

Users can load a (.species) file and this will append the information into their existing table.

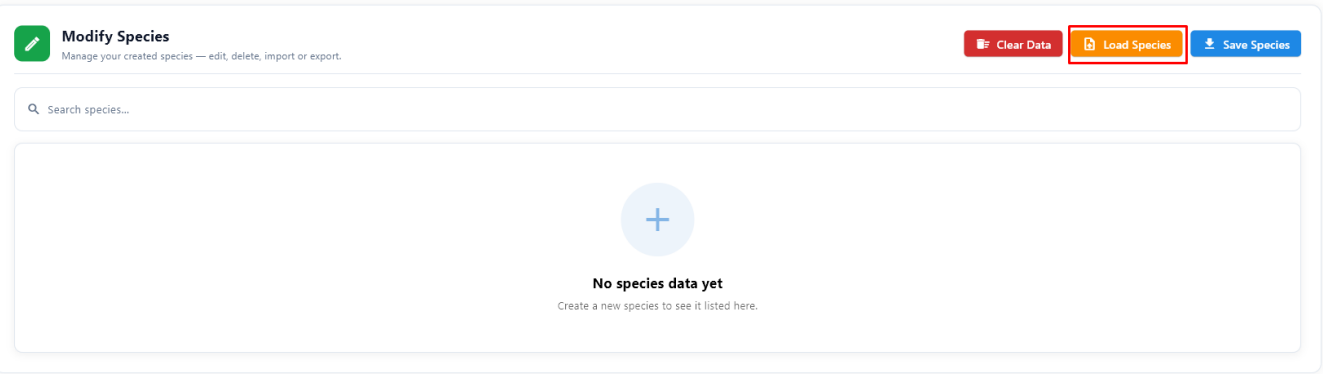


Figure 45. Click on load species button

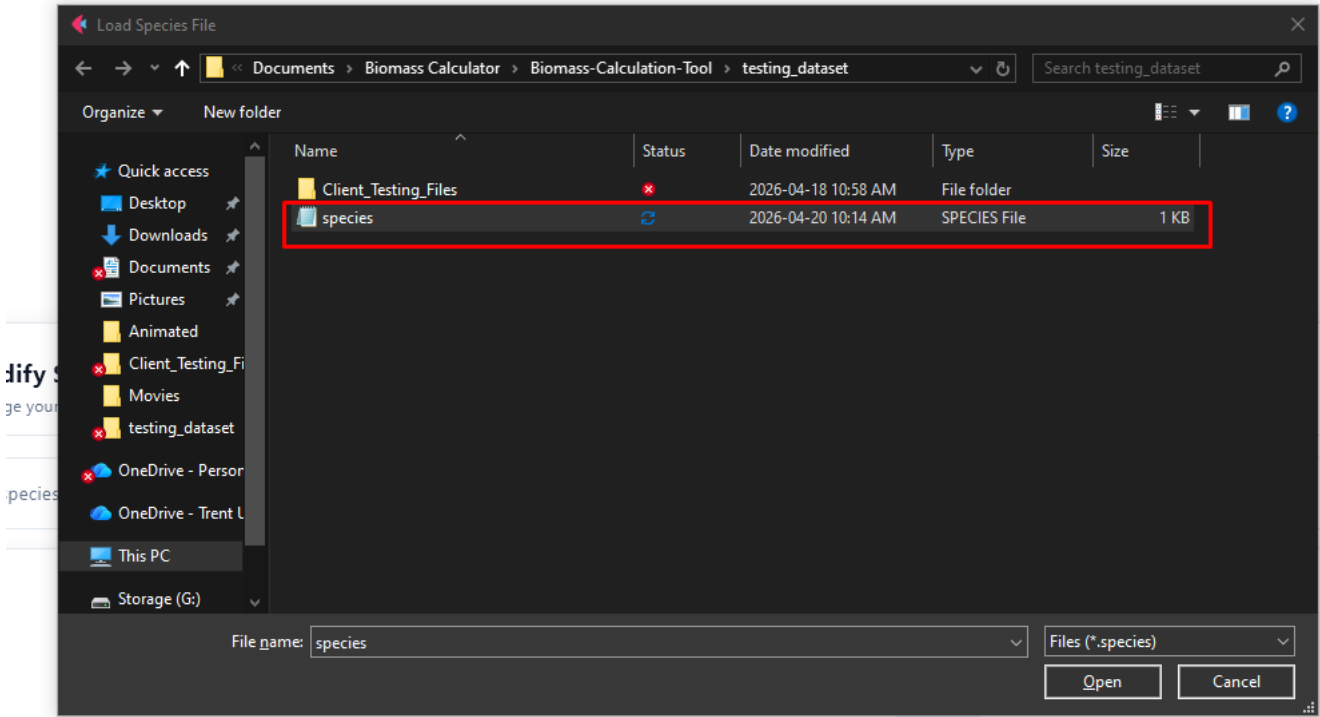


Figure 46. Select your .species file that you wish to import

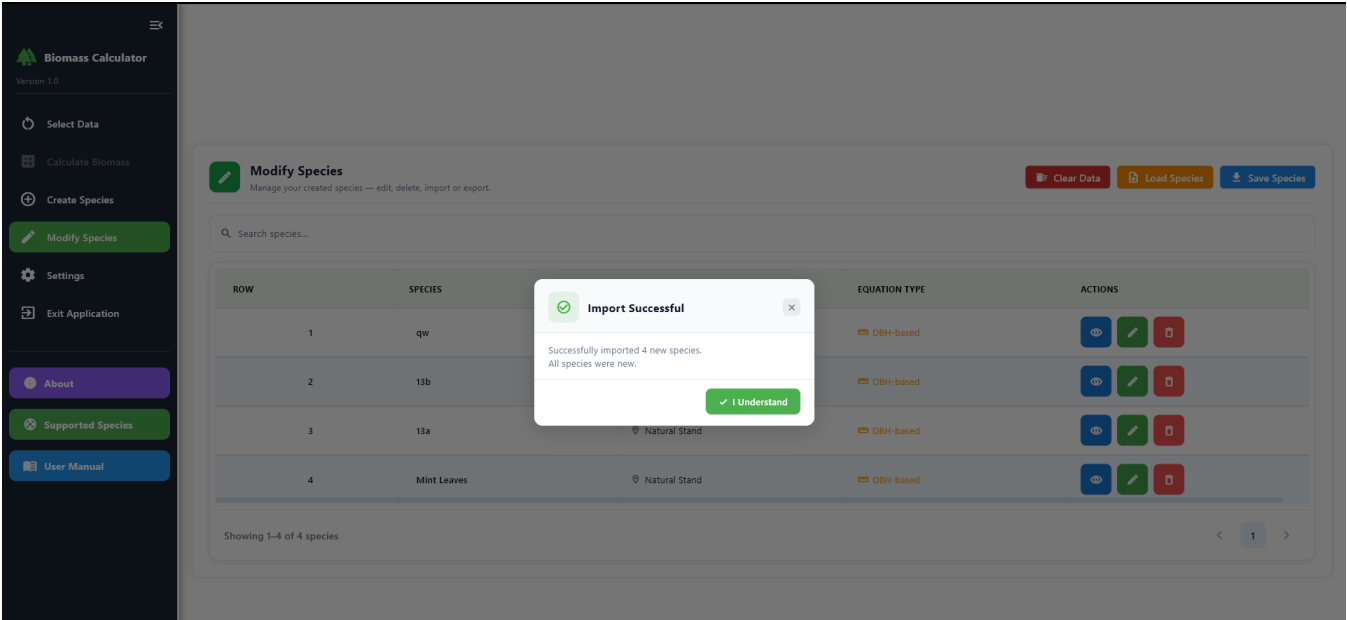
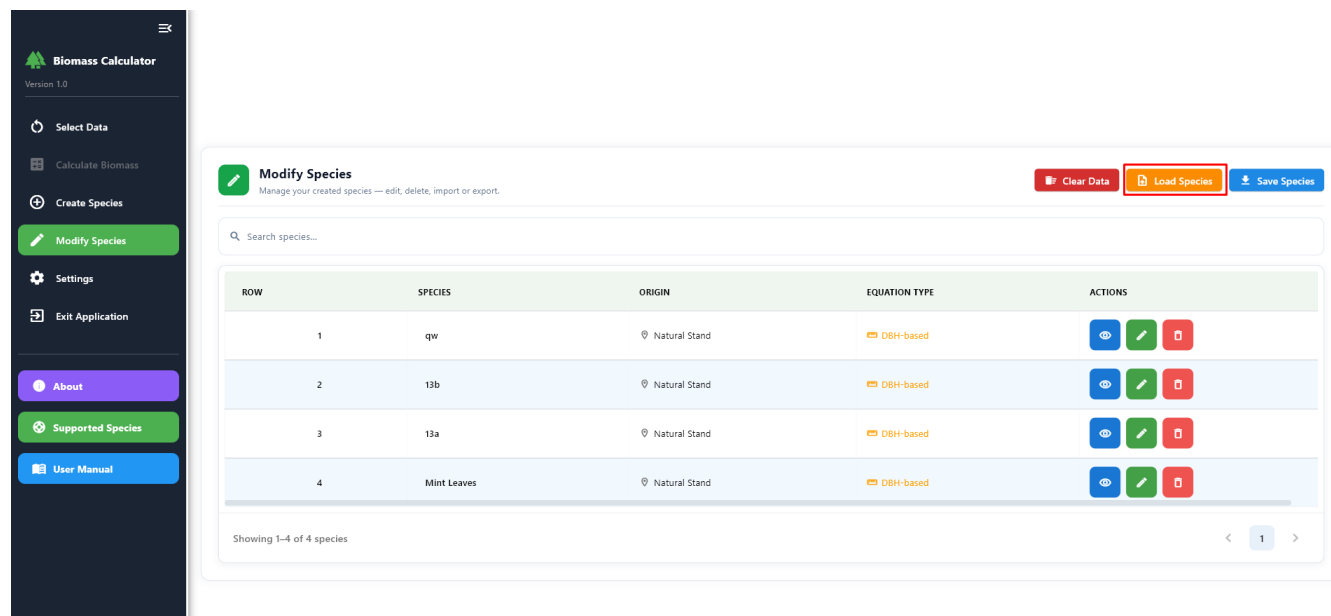


Figure 47. Successfully imported loaded species

7.f (i) – Load Species (Checking for duplicates)

Loading the species from a .species file will append to your existing table in other words it will merge your data with the imported or loaded species data.

Note: When you load a .species file, the data is merged with your existing table rather than replacing it. The program compares the current data with the imported data and adds only new, non-duplicate rows to avoid duplication.



Biomass Calculator
Version 1.0

- Select Data
- Calculate Biomass
- Create Species
- Modify Species**
- Settings
- Exit Application
- About
- Supported Species
- User Manual

Modify Species
Manage your created species — edit, delete, import or export.

Clear Data Load Species Save Species

Search species...

ROW	SPECIES	ORIGIN	EQUATION TYPE	ACTIONS
1	qw	Natural Stand	DBH-based	[Edit] [Add] [Delete]
2	13b	Natural Stand	DBH-based	[Edit] [Add] [Delete]
3	13a	Natural Stand	DBH-based	[Edit] [Add] [Delete]
4	Mint Leaves	Natural Stand	DBH-based	[Edit] [Add] [Delete]

Showing 1–4 of 4 species

Figure 48. Selecting “Load Species” button

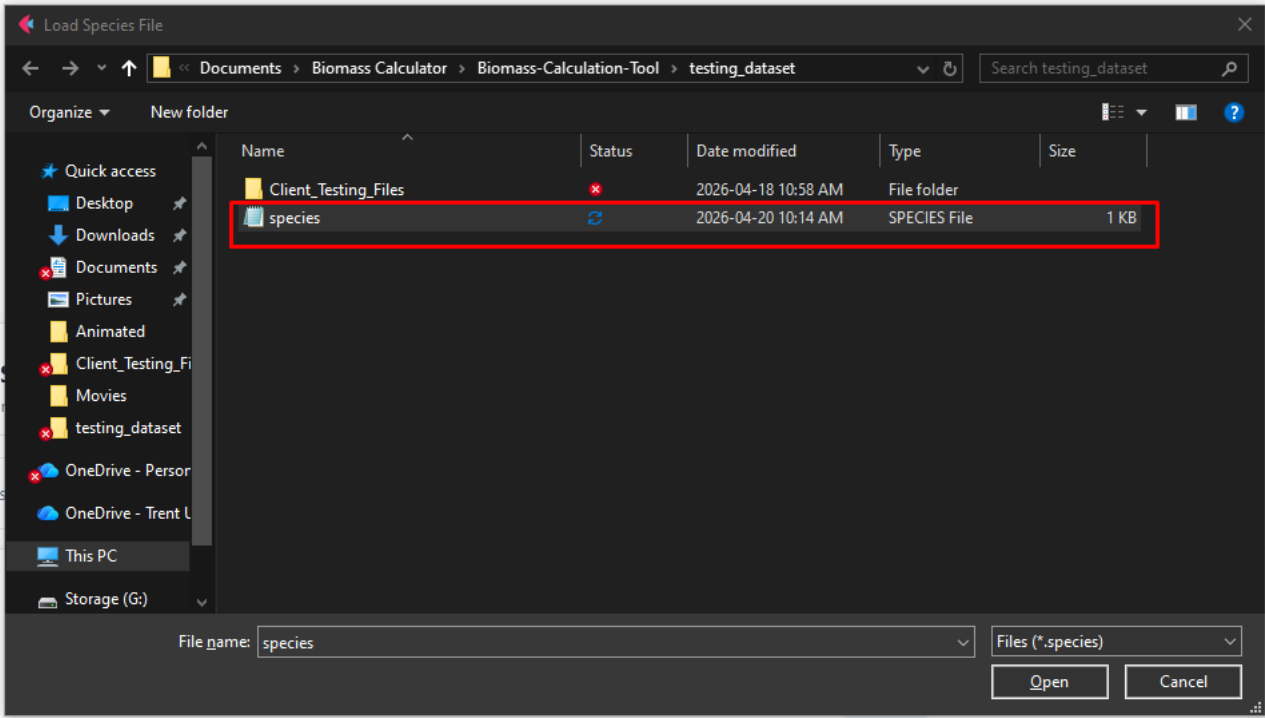


Figure 49. Selecting a .species file

After importing the same dataset used earlier, the program will not allow for the new data to be appended to the current user species data as it contains duplicates.

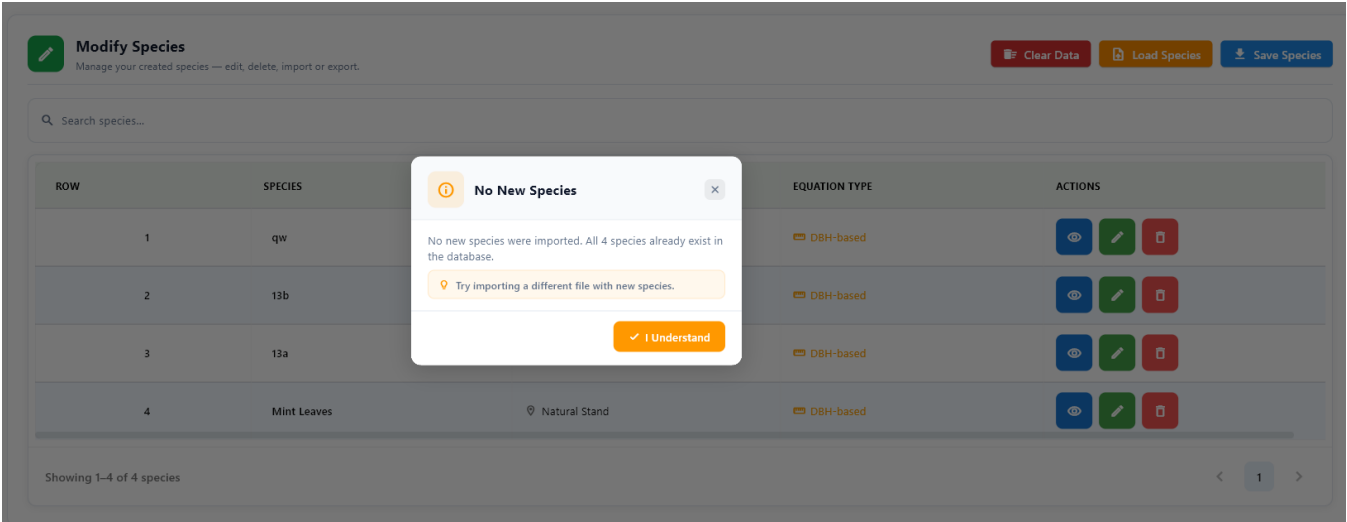


Figure 50. No new species dialog

7.f (ii) – Load Species (Appending data [non-duplicated])

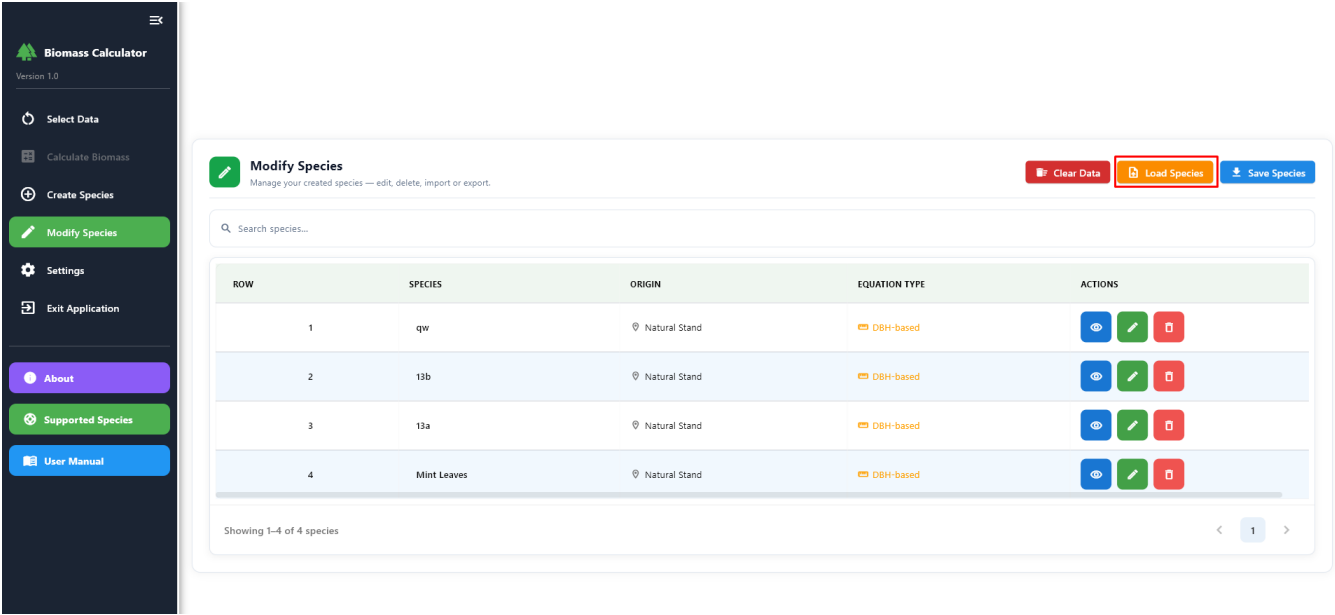
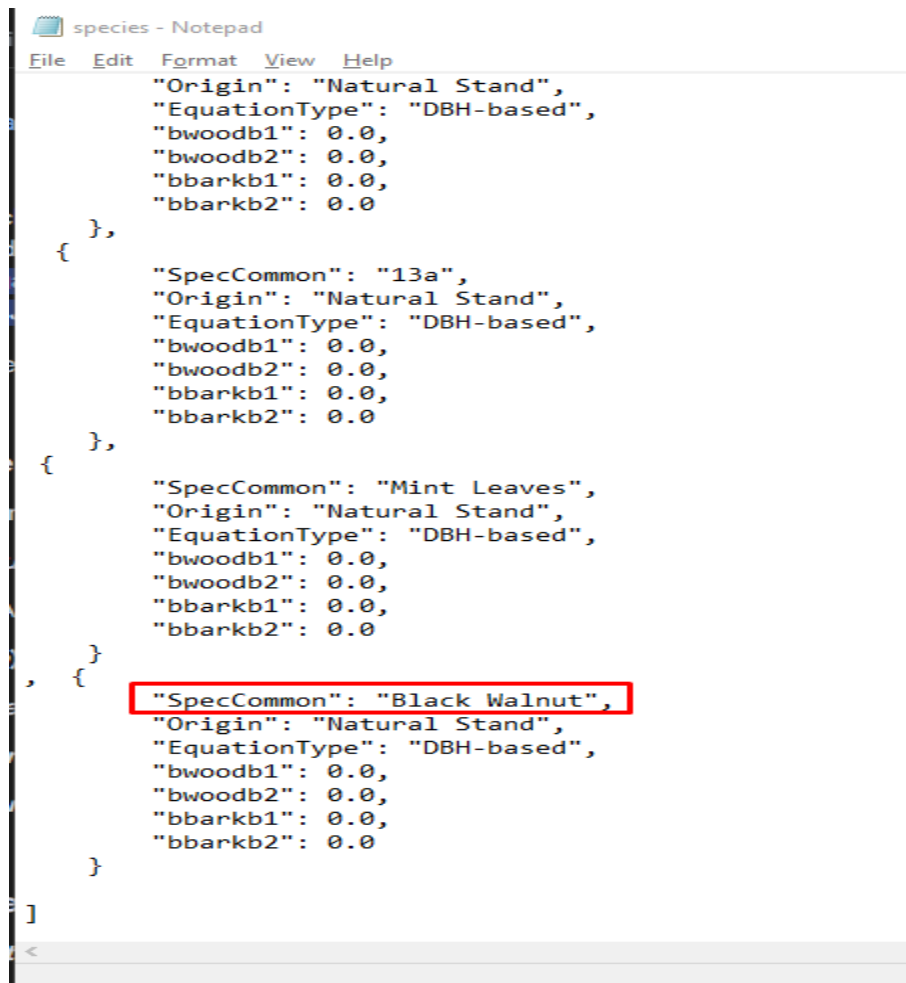


Figure 51. Selecting “Load Species” button



```

species - Notepad
File Edit Format View Help
{
  "Origin": "Natural Stand",
  "EquationType": "DBH-based",
  "bwoodb1": 0.0,
  "bwoodb2": 0.0,
  "bbarkb1": 0.0,
  "bbarkb2": 0.0
},
{
  "SpecCommon": "13a",
  "Origin": "Natural Stand",
  "EquationType": "DBH-based",
  "bwoodb1": 0.0,
  "bwoodb2": 0.0,
  "bbarkb1": 0.0,
  "bbarkb2": 0.0
},
{
  "SpecCommon": "Mint Leaves",
  "Origin": "Natural Stand",
  "EquationType": "DBH-based",
  "bwoodb1": 0.0,
  "bwoodb2": 0.0,
  "bbarkb1": 0.0,
  "bbarkb2": 0.0
},
{
  "SpecCommon": "Black Walnut",
  "Origin": "Natural Stand",
  "EquationType": "DBH-based",
  "bwoodb1": 0.0,
  "bwoodb2": 0.0,
  "bbarkb1": 0.0,
  "bbarkb2": 0.0
}
]

```

Figure 52. Added “Black Walnut” into the. species file

Here only 1 of the species was imported because the other three species were already existing with the current users table, so the program did not allow those to be imported but only “Black Walnut”.

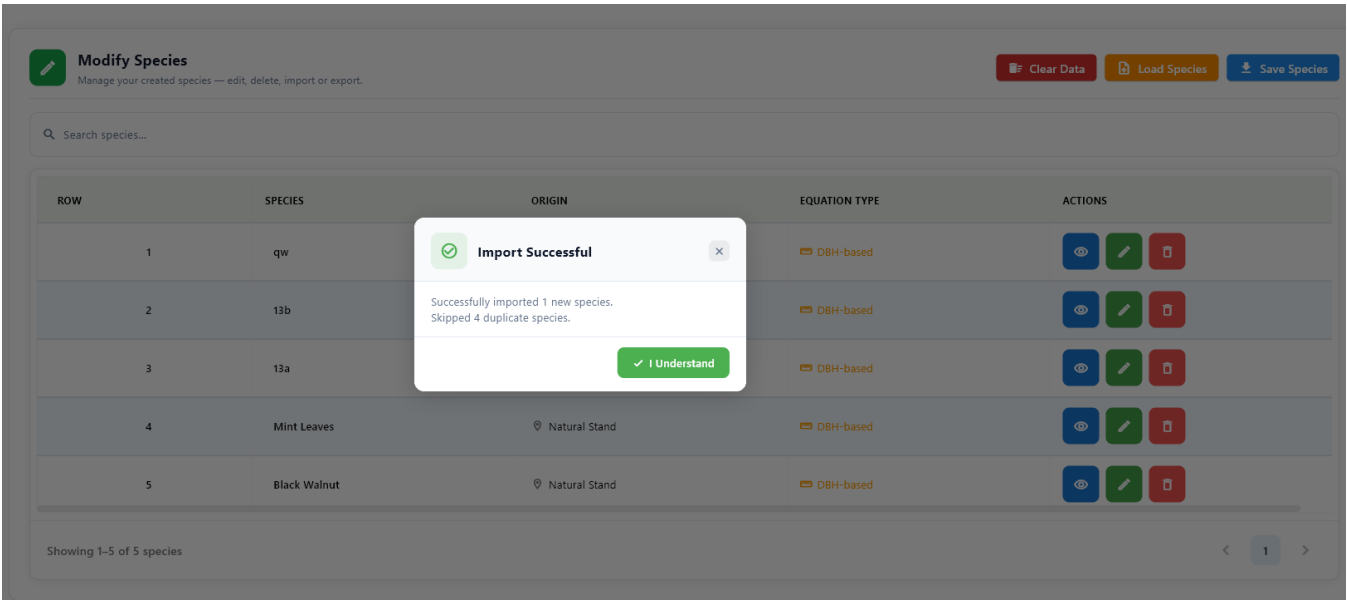


Figure 53. Import successful

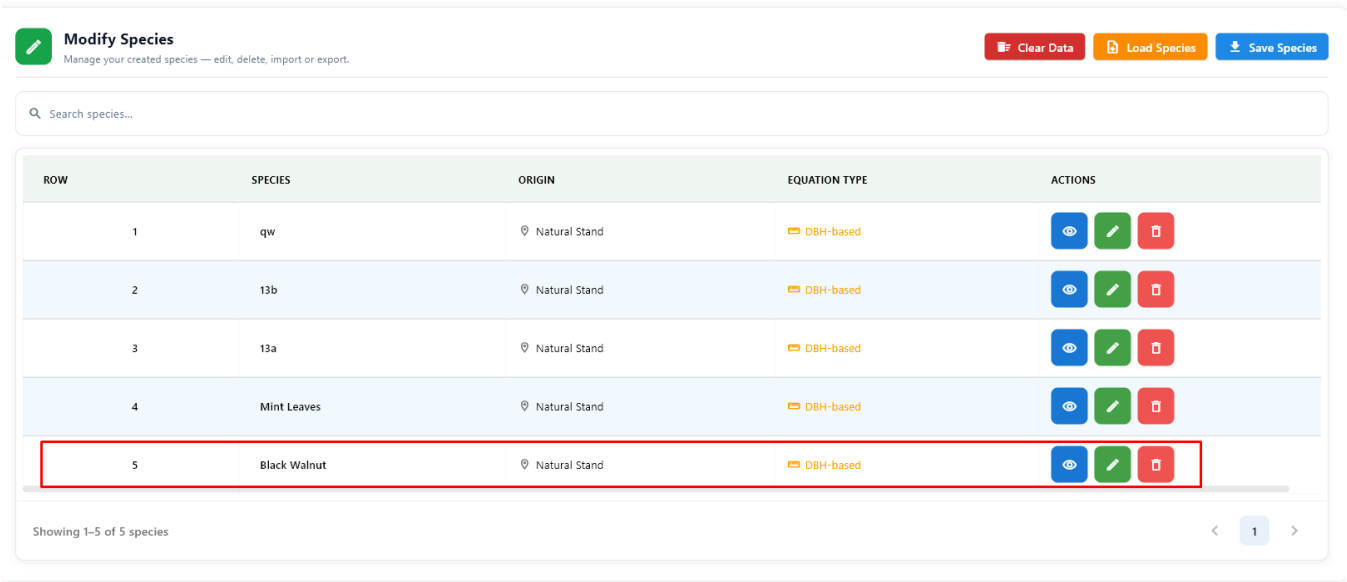


Figure 54. “Black Walnut” appended successfully

7.f (iii) – Load Species (Error)

If the program detects that the. species files have syntax errors; it will then display this output to the user.

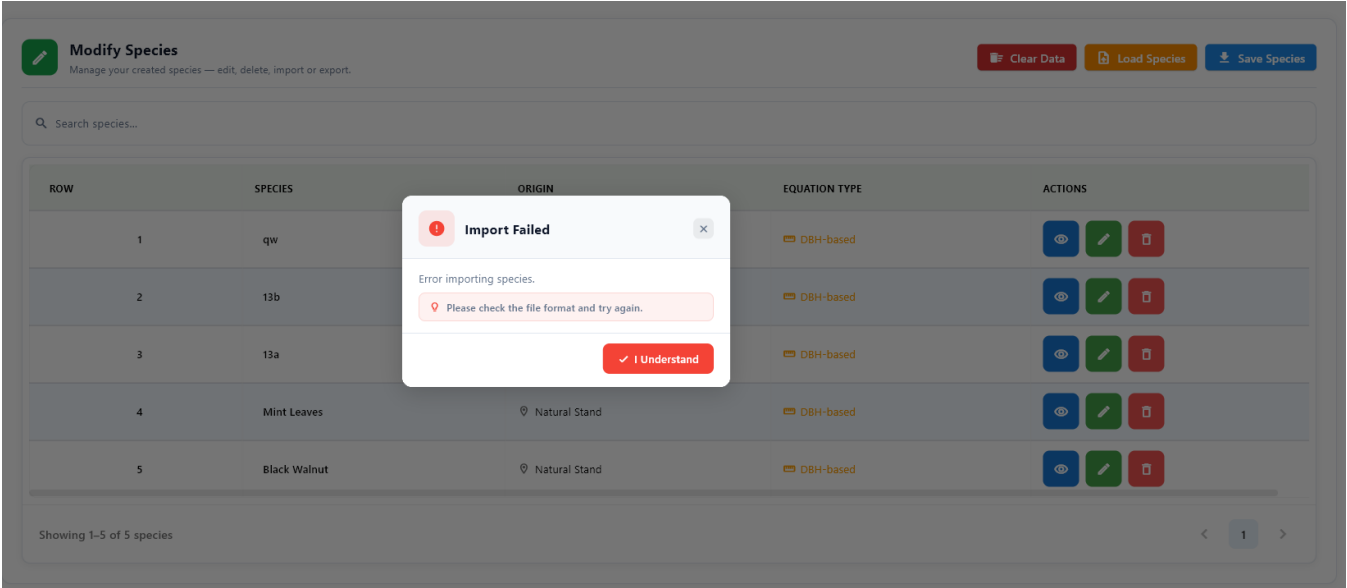


Figure 55. Error Dialog

7.g – Save Species

The program allows you to save your users created species as a. species file so you can share it with others or just save it as a backup file.



Figure 56. “Save Species” button is clicked

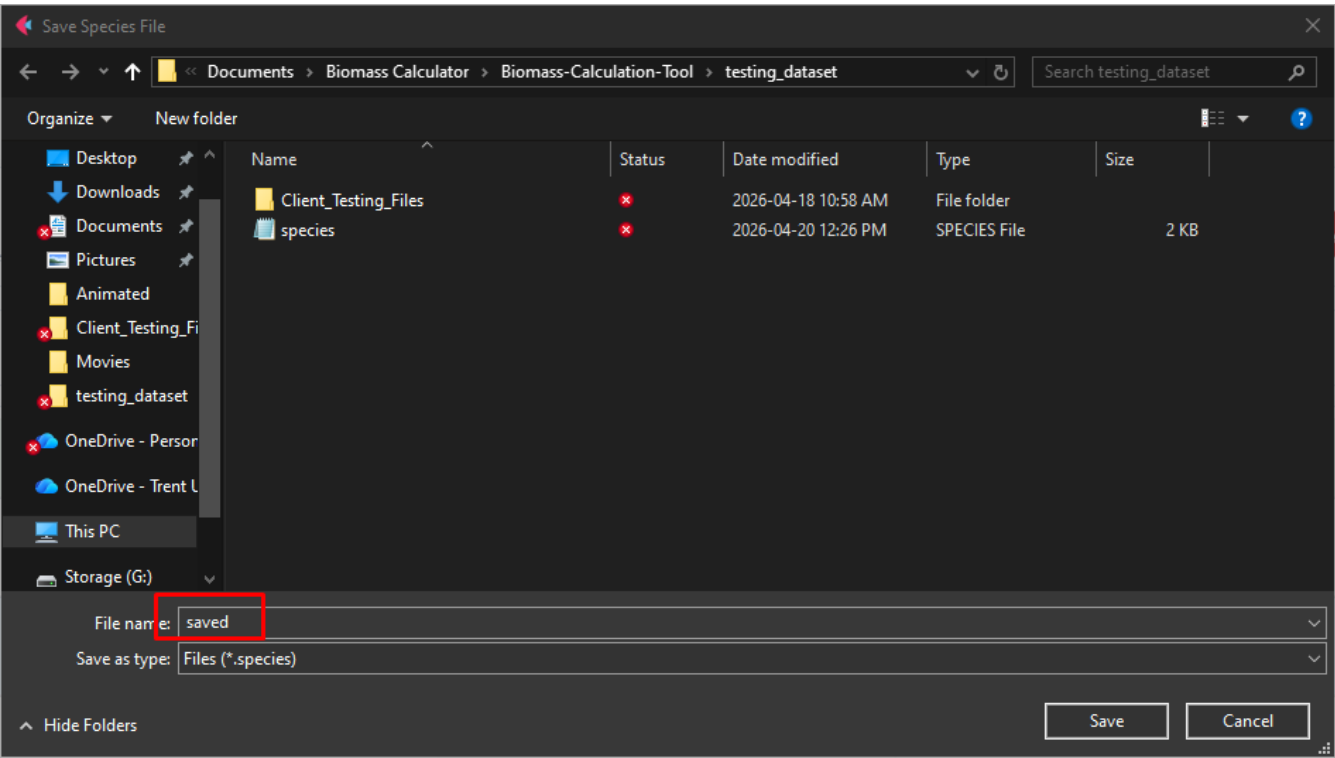


Figure 57. Saved as a. species file name called “saved”.

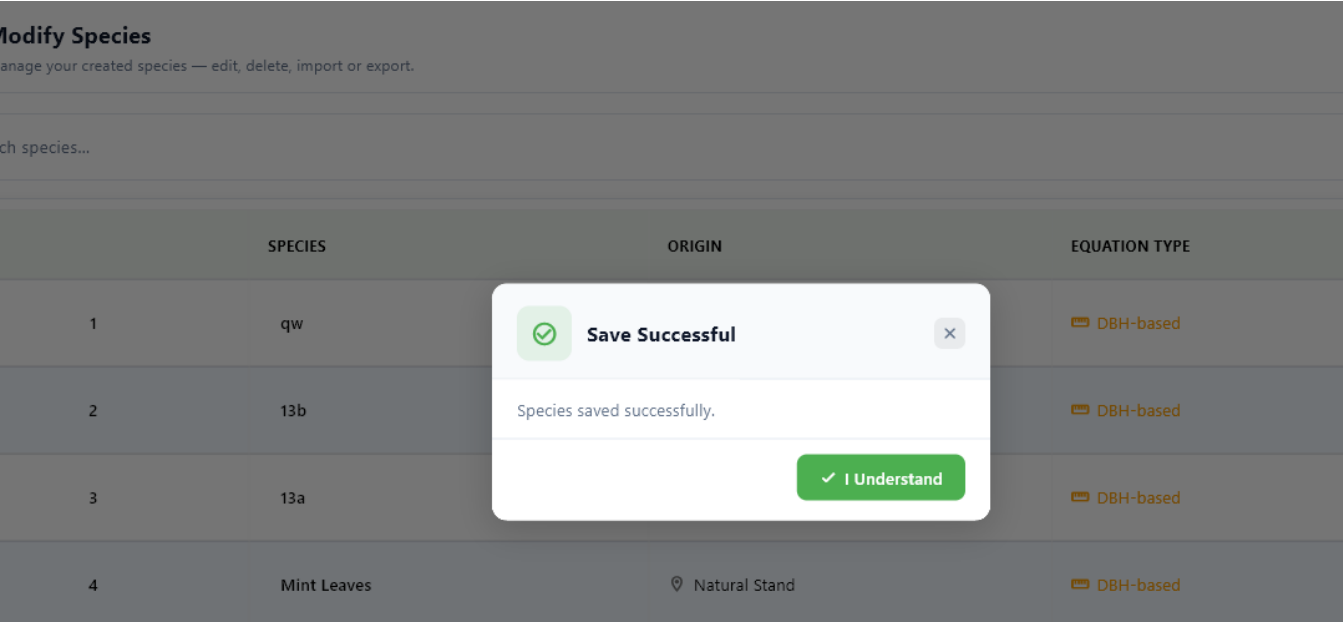


Figure 58. Save was successful

8) Settings

To change any settings, open the “Settings” menu by clicking on the Settings button on the left sidebar.

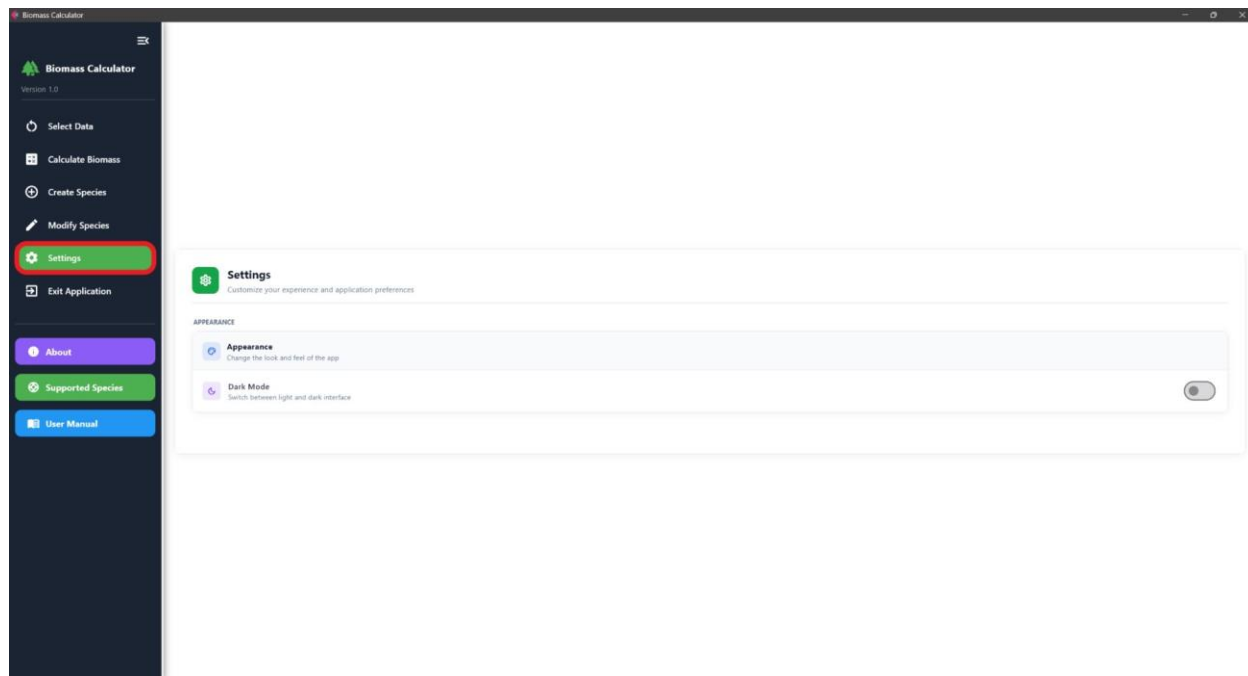


Figure 59. The [Settings] button is highlighted.

The only setting currently is to change the theme from Light to Dark.

9) Exit Application

To exit the application just click on the “Exit Application” button on the left sidebar.

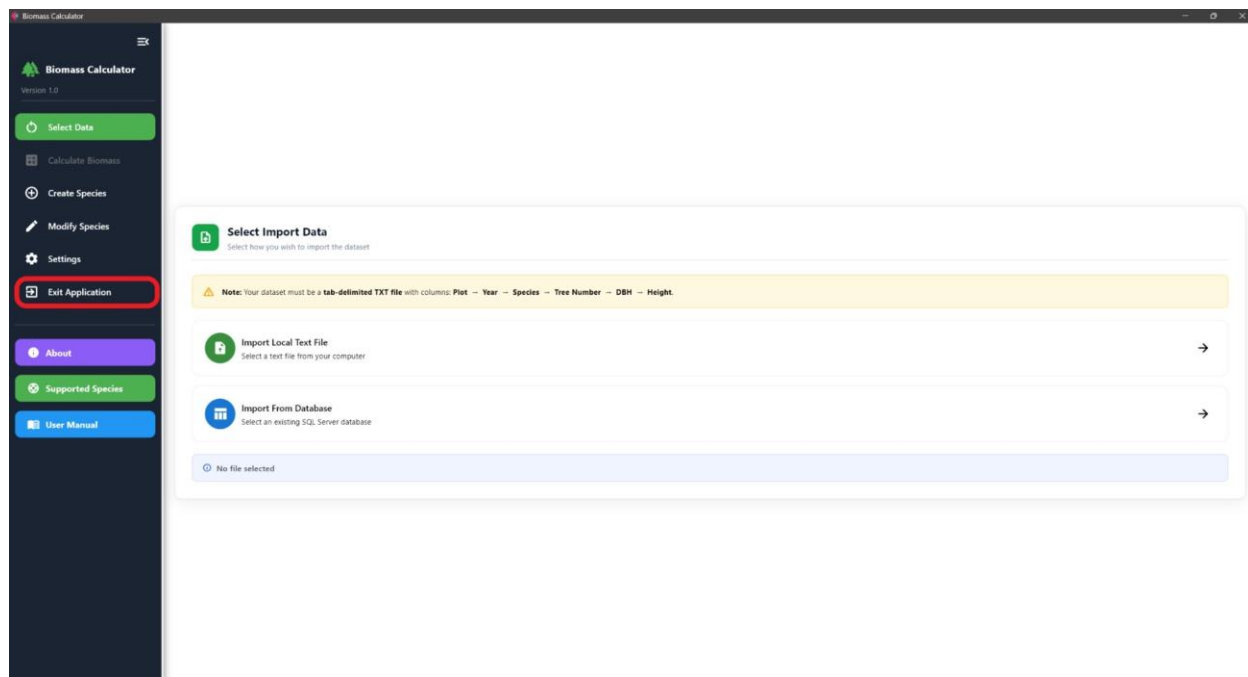


Figure 60. The [Exit Application] button is highlighted.

Then click on the “Exit Anyway” button on the menu that shows up.

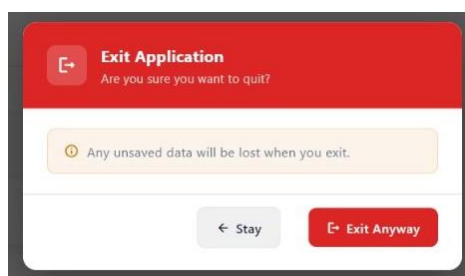


Figure 44. The Confirm Exit menu.

Alternatively, just click on window  in the top right.

10) Log

When you are done with your session and exit the calculator, you can go to the “log” directory (which is in the same directory as the program) to find dated text files with timestamped records of your activities for that session and all previous sessions.

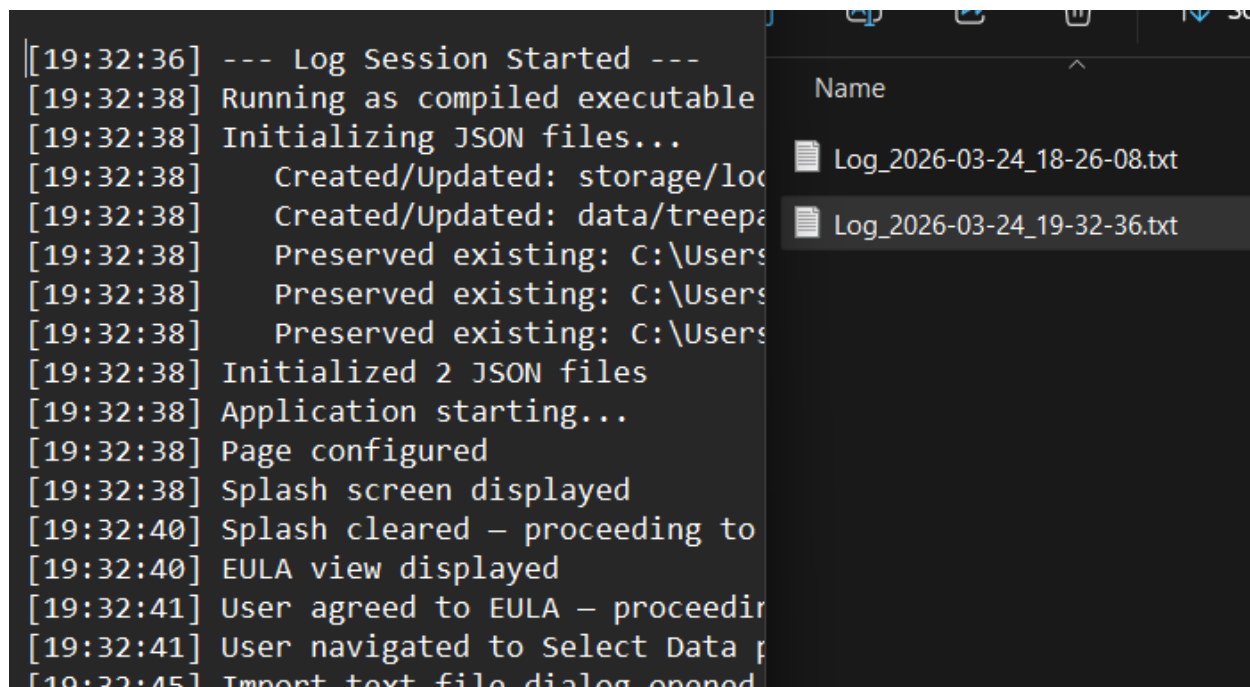


Figure 61. The window X button is highlighted.

